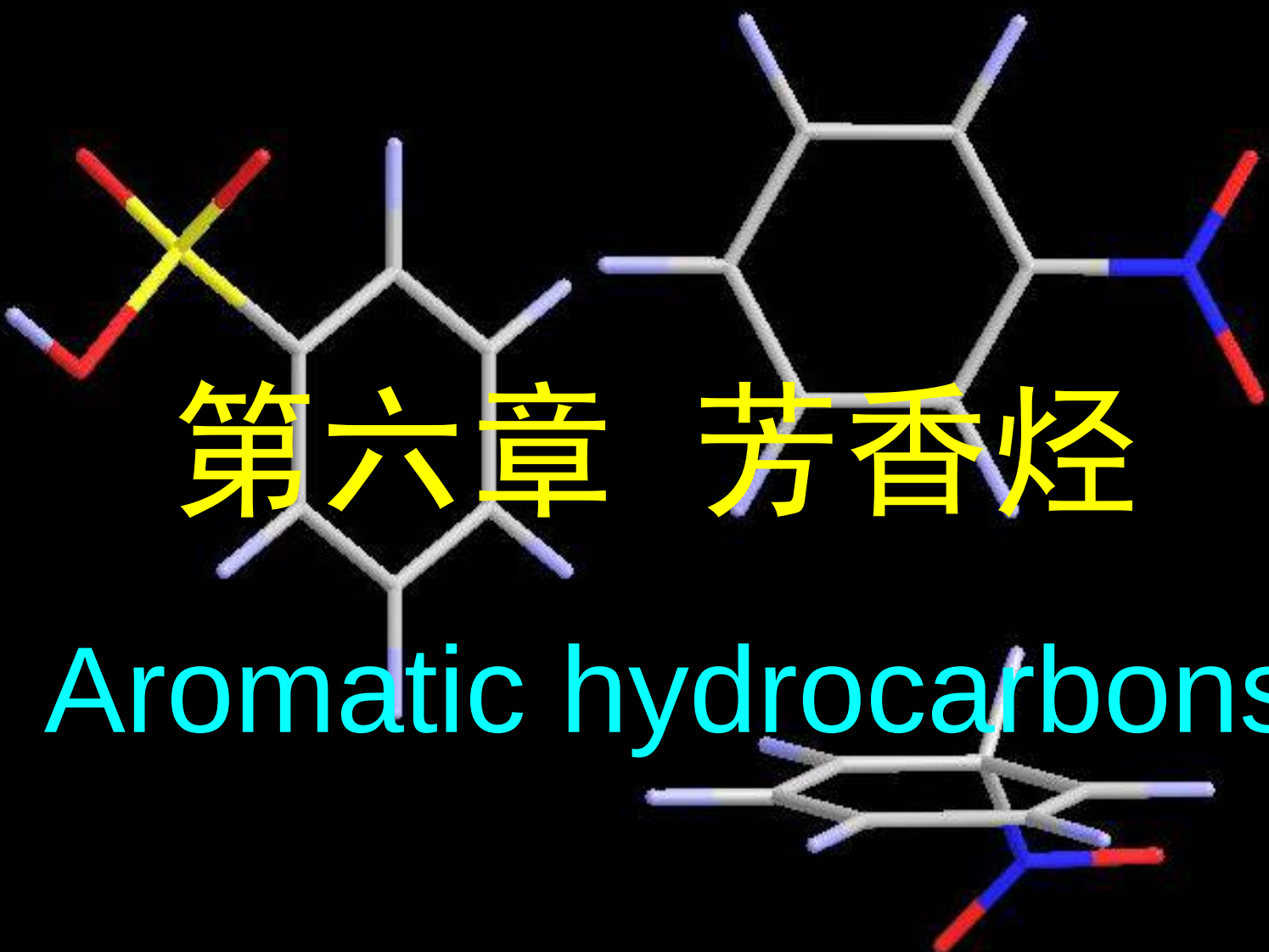


The background features several molecular structures. On the left, a benzene ring is substituted with a sulfonamide group (-SO₂NH₂) and a methyl group. On the right, a benzene ring is substituted with a methyl group and a dimethylamino group (-N(CH₃)₂).

第六章 芳香烃

Aromatic hydrocarbons

厦门大学有机化学B

CONTENT

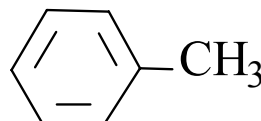
- 1 芳香烃的分类、结构及命名
- 2 芳香烃的物理性质(自学)
- 3 芳香烃的化学性质
- 4 苯环上亲电取代的定位规律
- 5 稠环芳烃

6.1 芳香烃的分类、结构及命名

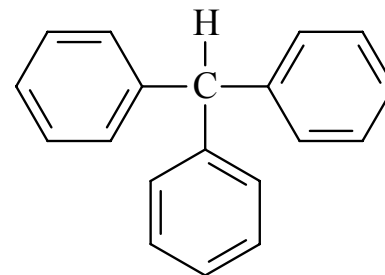
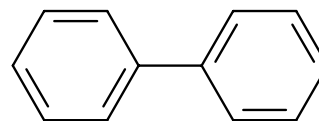
6.1.1 芳香烃的分类

芳香烃

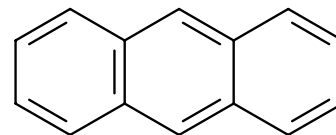
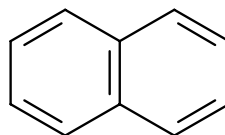
单环芳烃



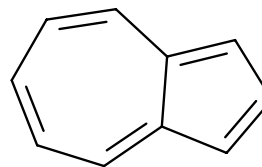
多环芳烃



稠环芳烃



非苯芳烃



不含苯环，但具有苯的特性“芳香性”的碳环化合物

6.1.2 苯的结构

1825年法拉第(Faraday)从鲸鱼脂肪中分离得到

苯的特征:

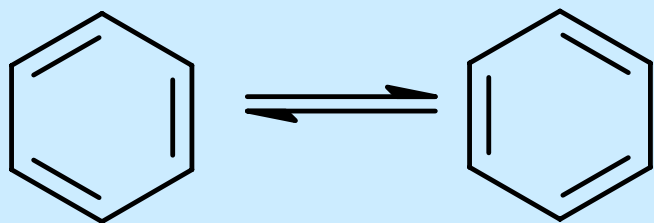
- 分子式 C_6H_6 , $C:H=1:1$, 不饱和度为4
- 具有特殊稳定性和对化学反应的惰性

芳香族化合物的特性 —— 芳香性

- 苯环易发生取代, 难发生加成反应
- 不易氧化
- 苯环具有特殊稳定性

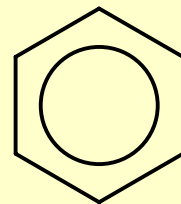
最为人们所普遍接受的结构

共轭环己三烯苯——Kekulé 苯



Kekulé 苯

Robinson 1925年
建议采用

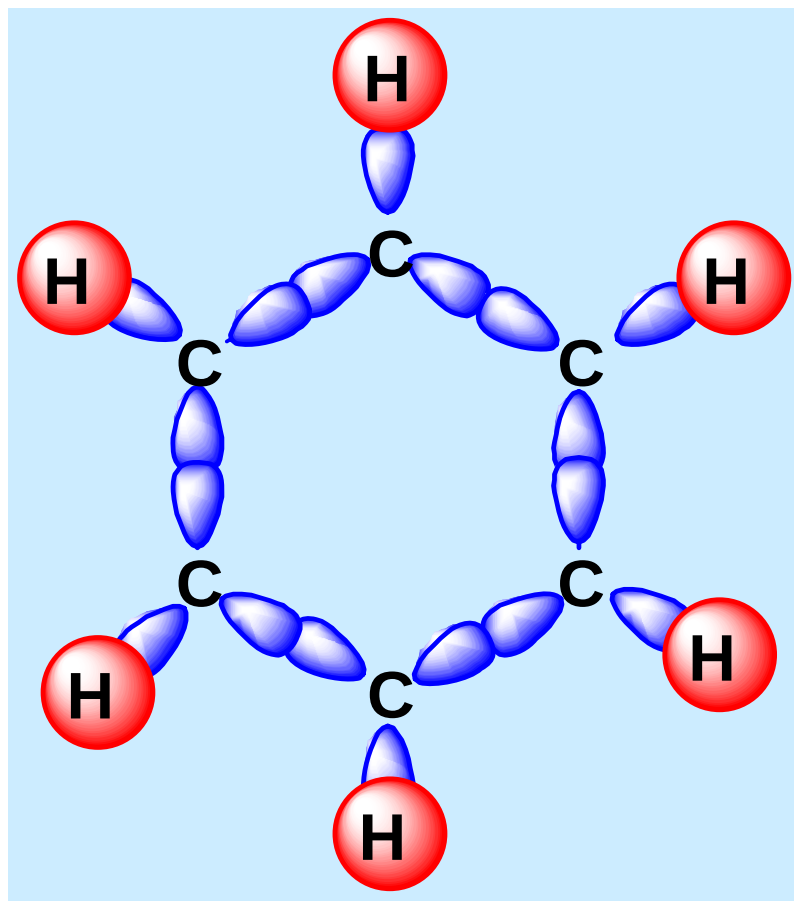


近代物理方法对苯结构的研究结果

- 苯的6个碳原子形成平面正六边形
- 碳-碳键长均等，为 $1.40 \text{ \AA}$
- 所有键角均为 120°

苯的分子轨道描述

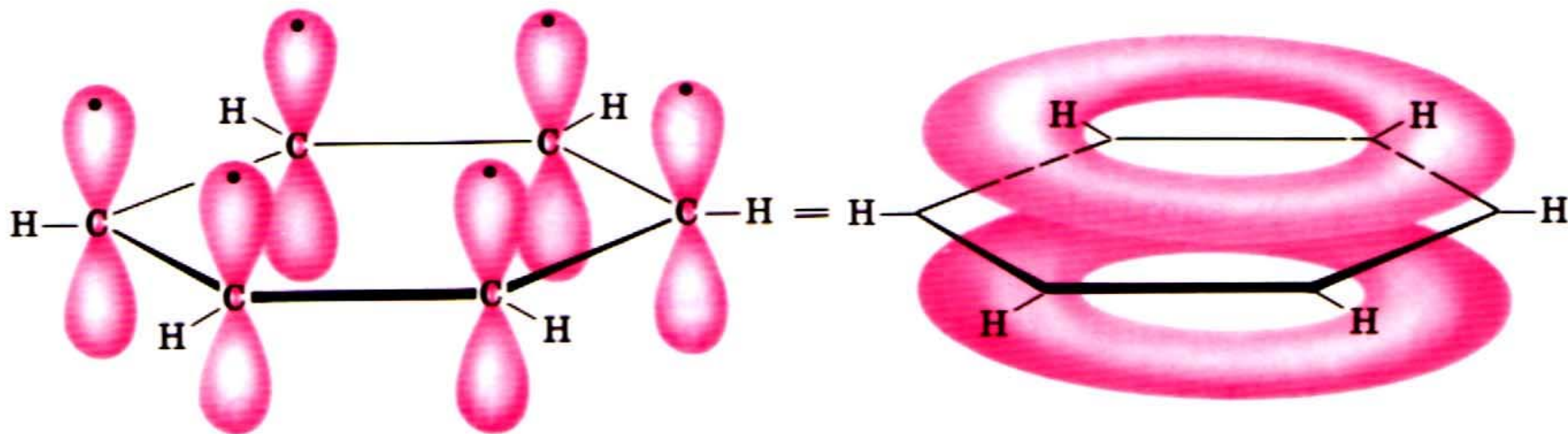
- 6个碳原子都是 sp^2 杂化



厦门大学有机化学B

苯的分子轨道描述

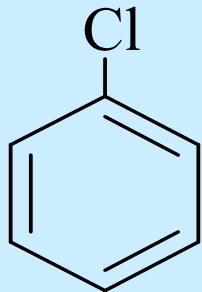
- 未参与杂化的 p 轨道带着一个电子垂直于环平面，形成一个闭合环状的大 π 键



6.1.3 苯及其衍生物的命名和异构

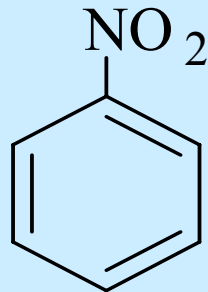
1. 单取代苯的命名

苯环作为母体：—X，—NO₂，—NO，简单烷基



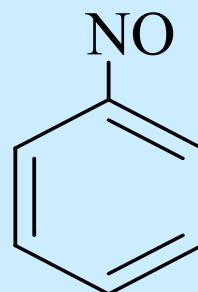
氯苯

Chlorobenzene



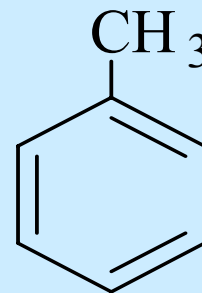
硝基苯

nitrobenzene



亚硝基苯

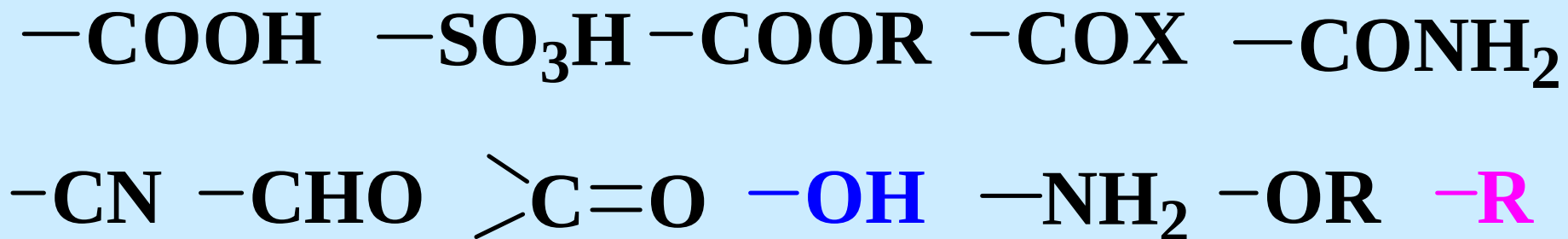
nitrosobenzene



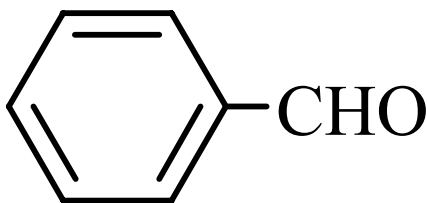
甲苯

methylbenzene
(Toluene)

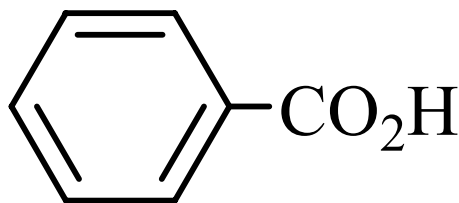
苯环作为取代基：**R**之前基团与苯环相连时



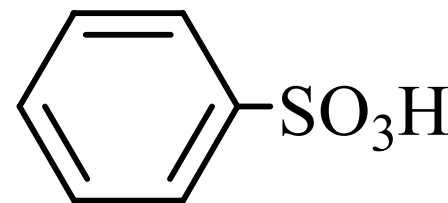
— OH, —NH₂与苯环一起构成母体“苯酚”、“苯胺”



苯甲醛



苯甲酸

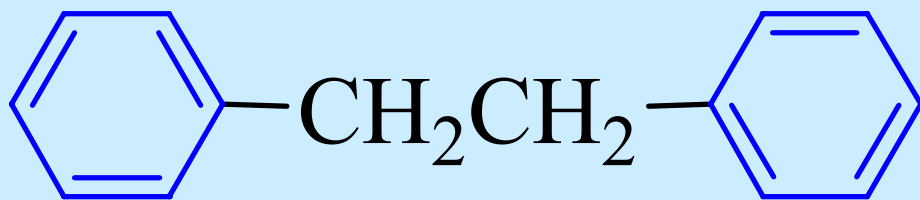


苯磺酸

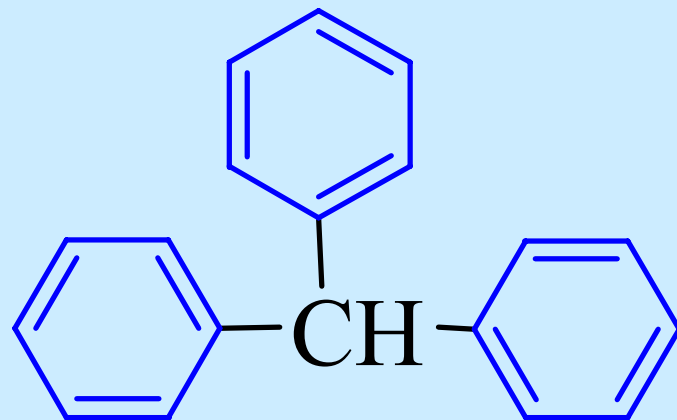
厦门大学有机化学B

苯环作为取代基

分子中有多个苯环，
以任一苯环为母体命名都有困难时



1,2-二苯基乙烷

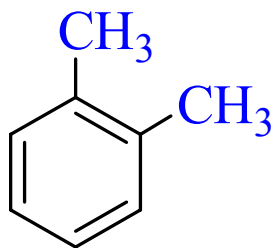


三苯基甲烷

2. 二取代苯的命名

- 2个取代基相同

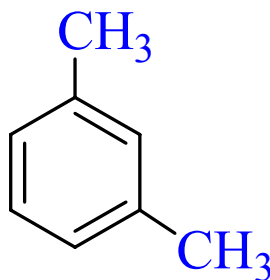
(1,2- *o*- 邻; 1,3- *m*- 间; 1,4- *p*- 对)



1,2-Dimethylbenzene
(*o*-Xylene)

ortho

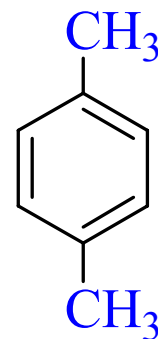
1,2-二甲苯
邻-二甲苯



1,3-Dimethylbenzene
(*meta*-Xylene)

meta

1,3-二甲苯
间-二甲苯

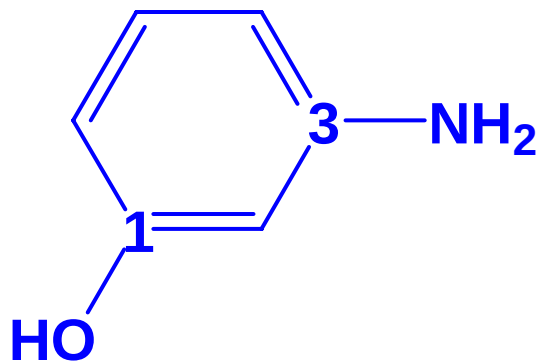


1,4-Dimethylbenzene
(*para*-Xylene)

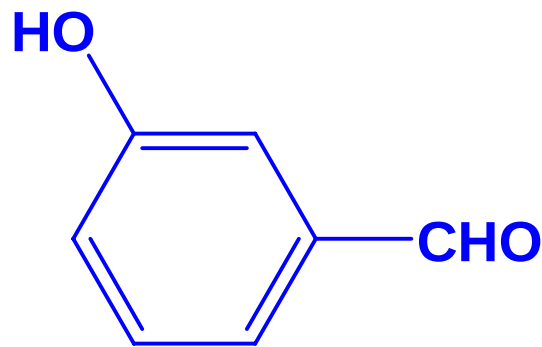
para

1,4-二甲苯
对-二甲苯

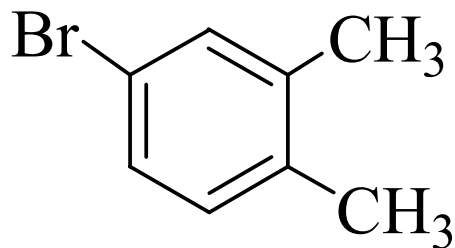
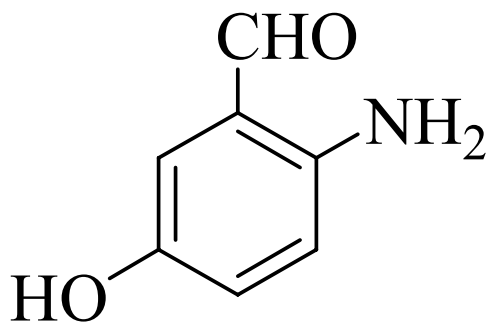
- 多取代苯(确定主官能团, 按最低系列原则编号)



3-氨基苯酚



3-羟基苯甲醛



6.3 芳香烃的化学性质

芳香亲电取代反应

加成反应

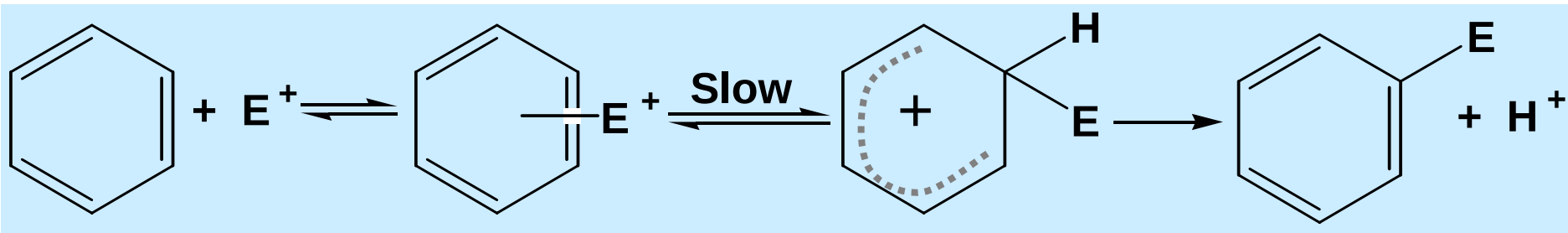
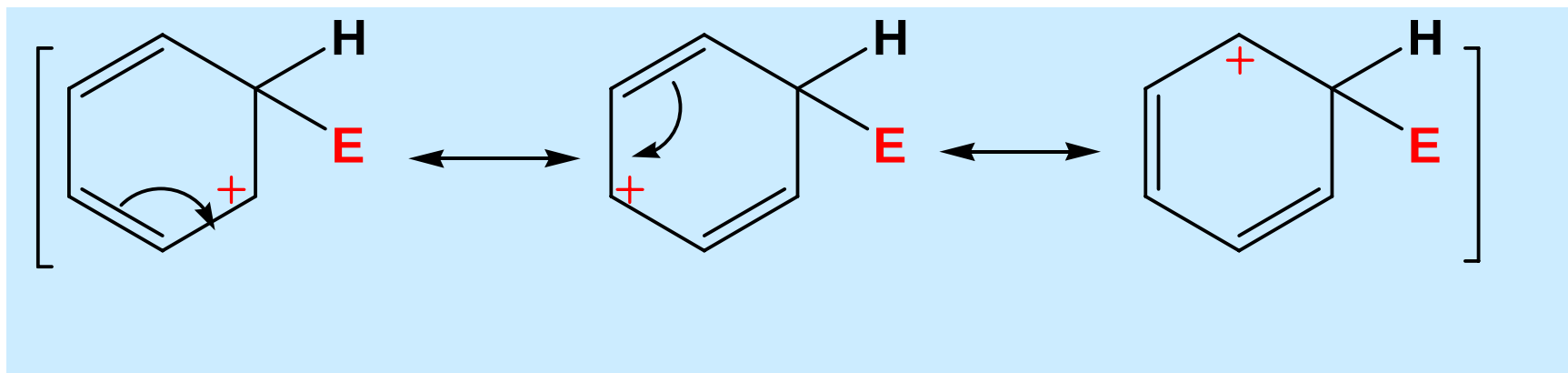
氧化反应

α -卤代反应

厦门大学有机化学B

6.3.1 芳香亲电取代反应

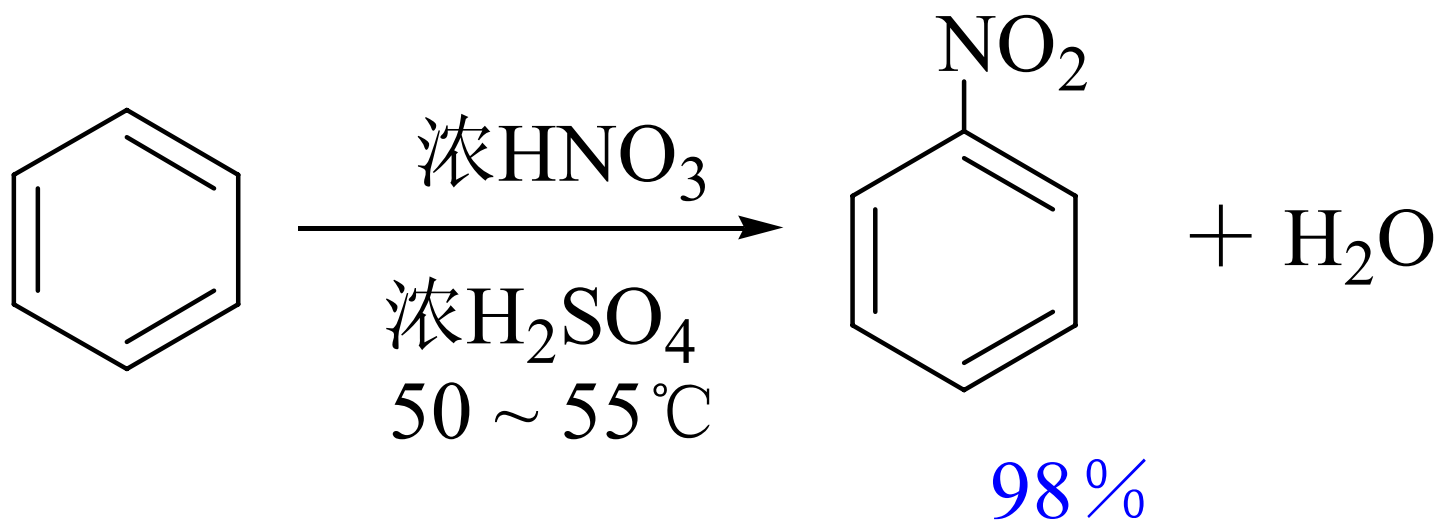
(Electrophilic Aromatic Substitution)



π 络合物

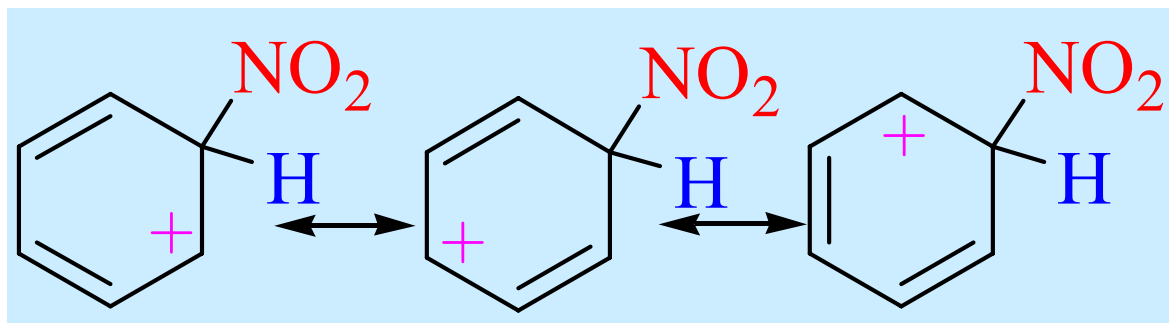
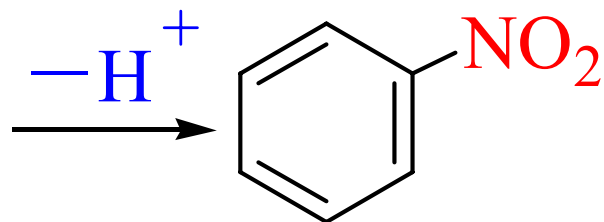
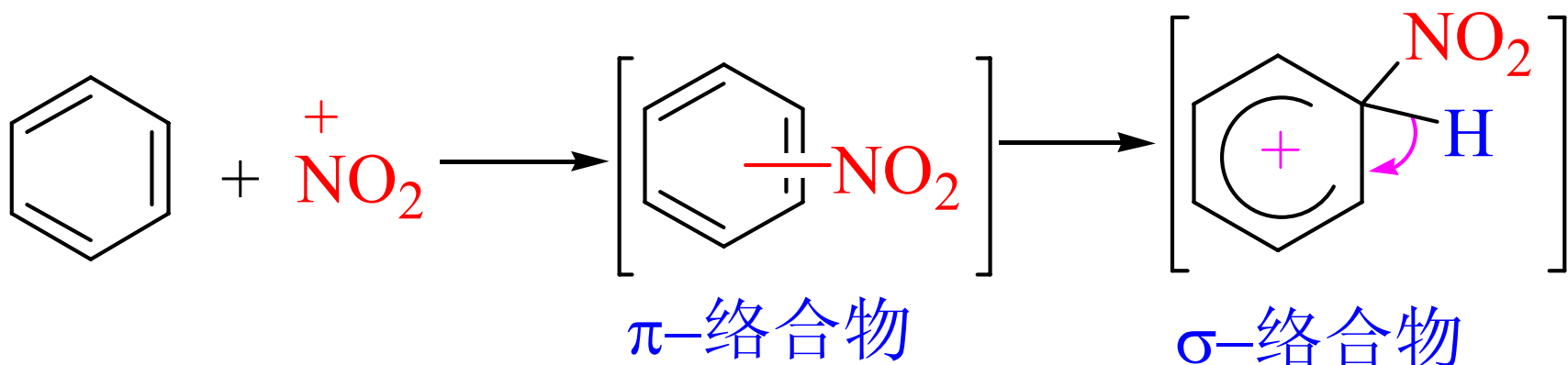
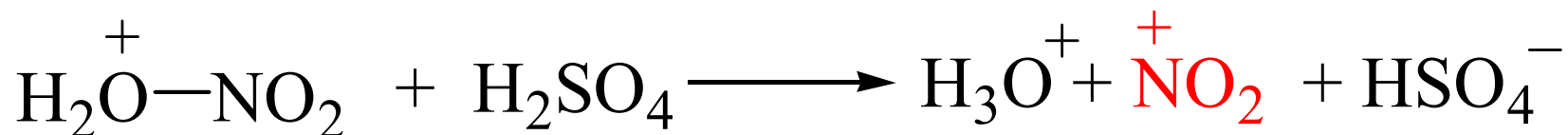
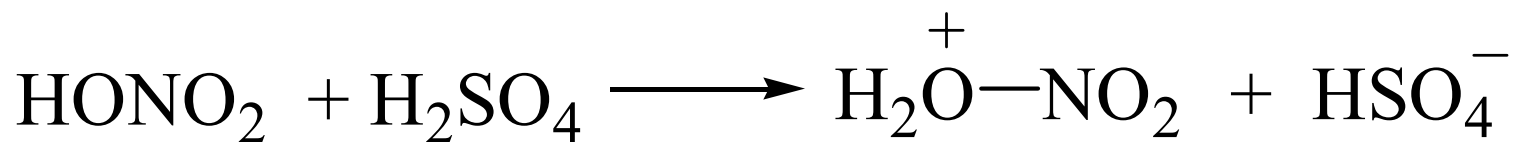
碳正离子
(σ 络合物)

1. 硝化(Nitration)

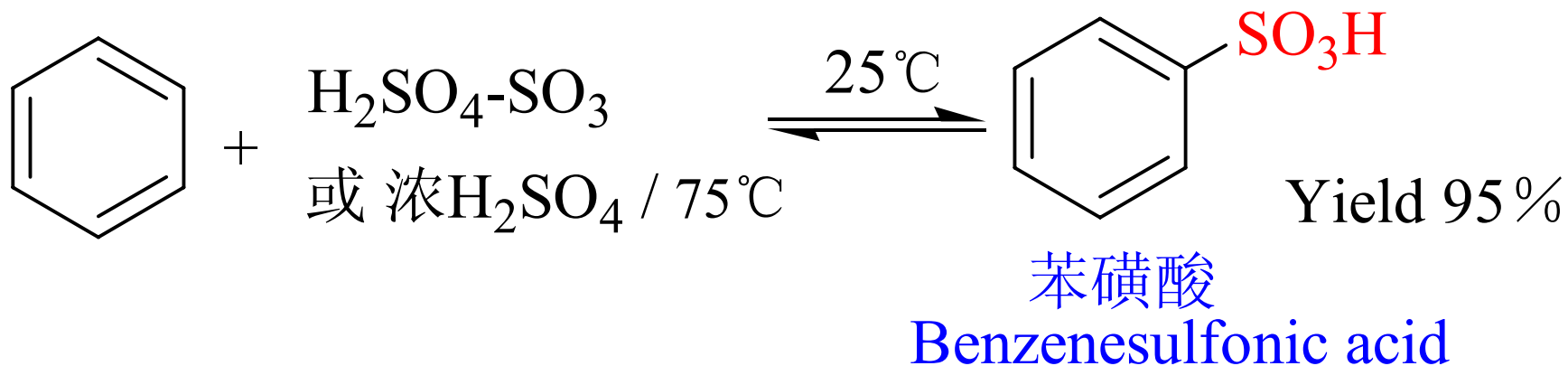


厦门大学有机化学B

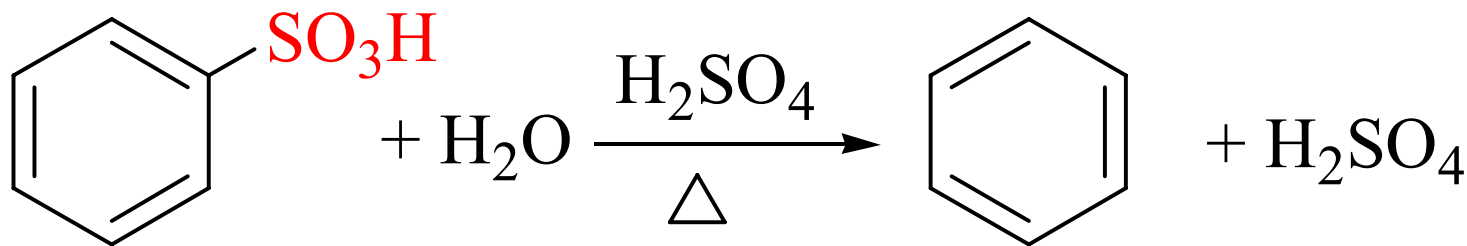
反应机理



2. 磺化反应(Sulfonation)

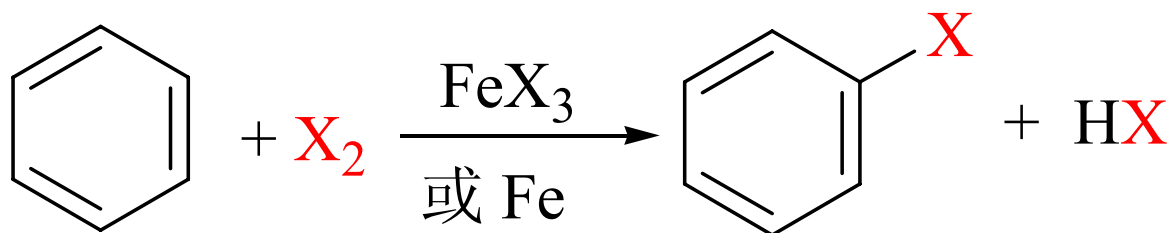


- 苯磺酸在稀酸溶液中加热，磺酸基水解脱落

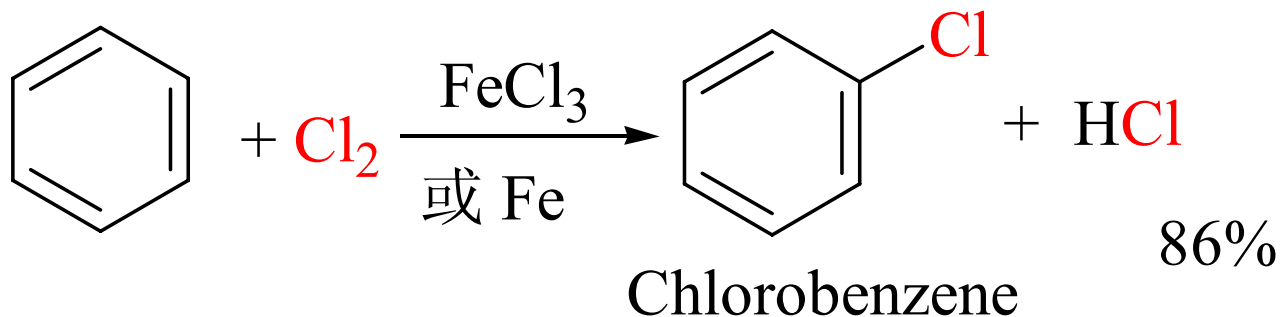
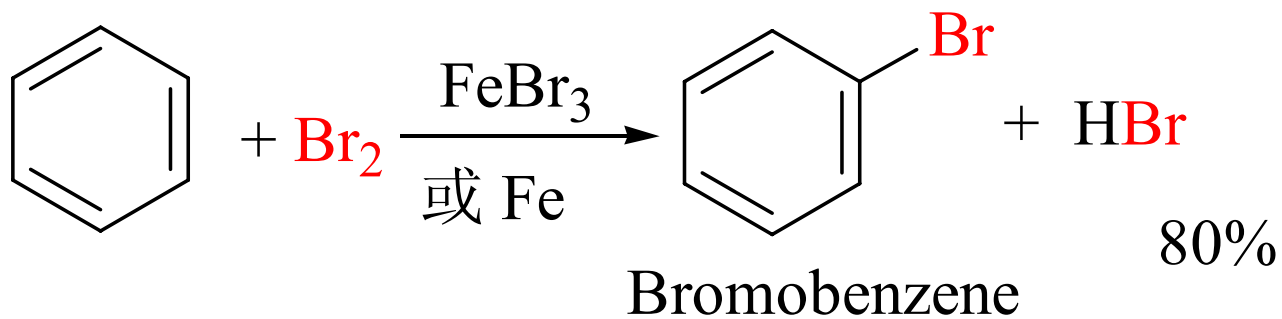


磺化反应具有重要用途

3. 卤化反应(Halogenation)

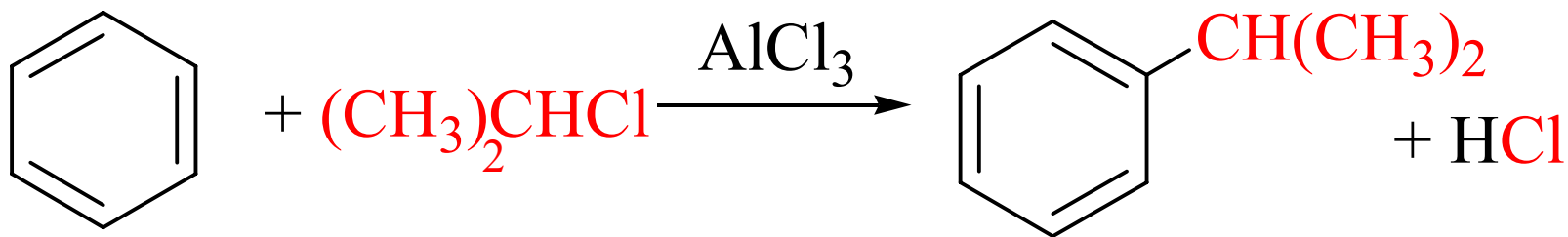
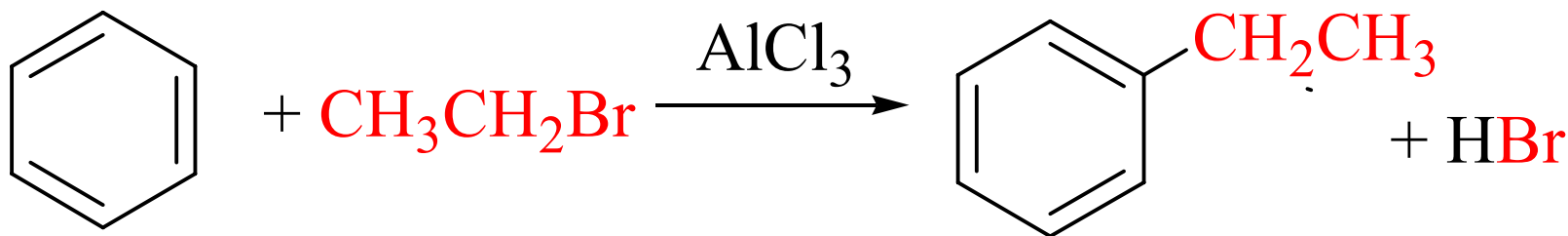


反应活性: $F_2 > Cl_2 > Br_2 > I_2$



4. 傅-克 (Friedel—Crafts) 反应

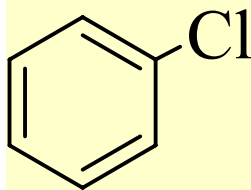
(A) 烷基化(Alkylation)



催化剂 Lewis 酸

FeCl_3 , SnCl_4 , ZnCl_2

HF , H_2SO_4 , H_3PO_4 , BF_3

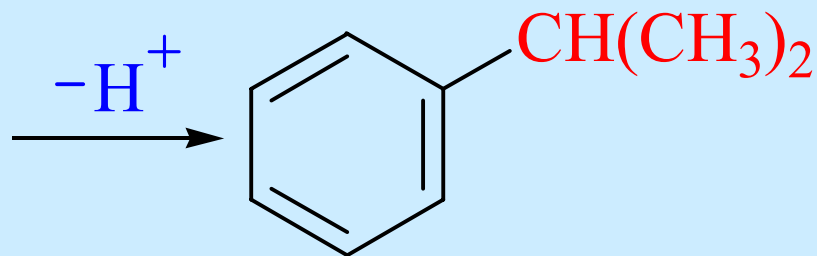
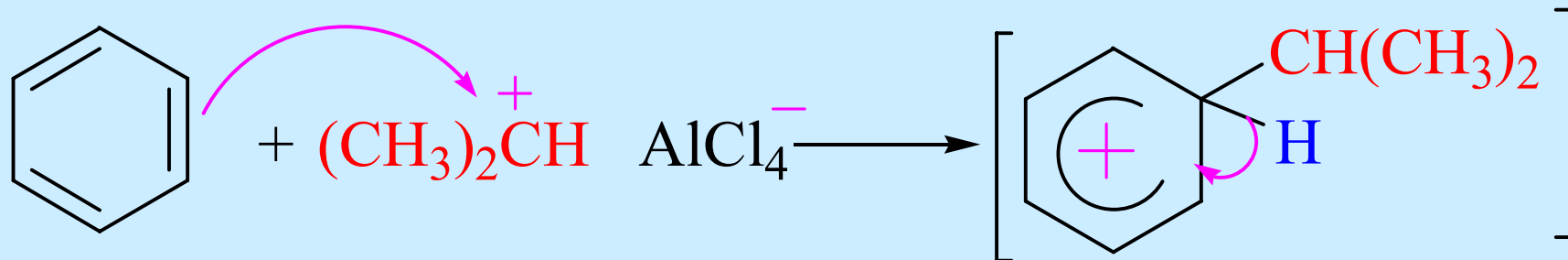
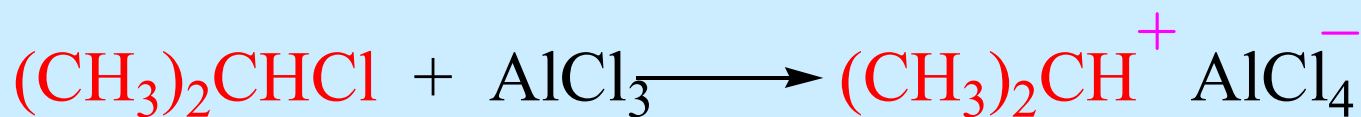


$\text{CH}_2=\text{CHCl}$

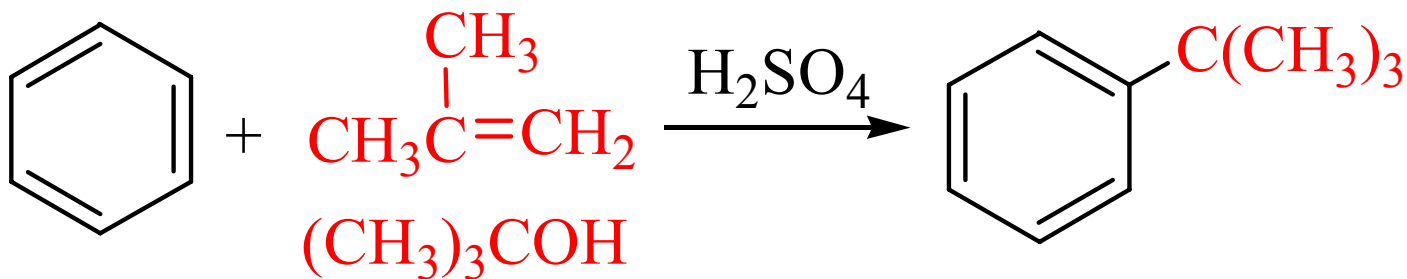
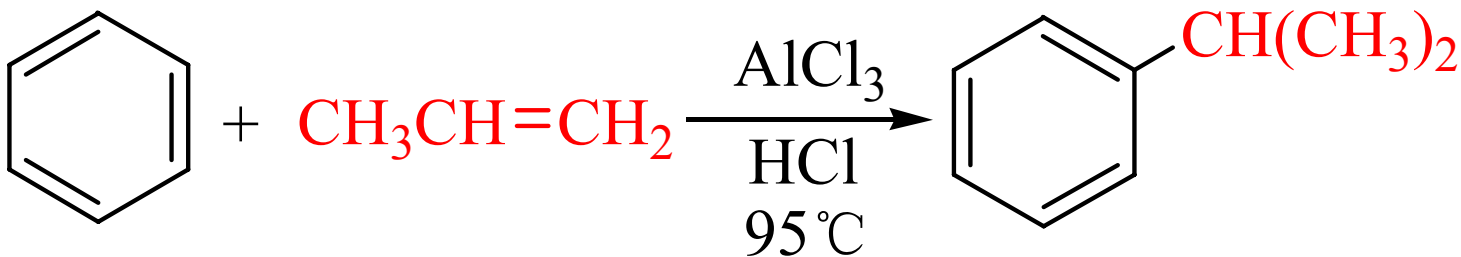
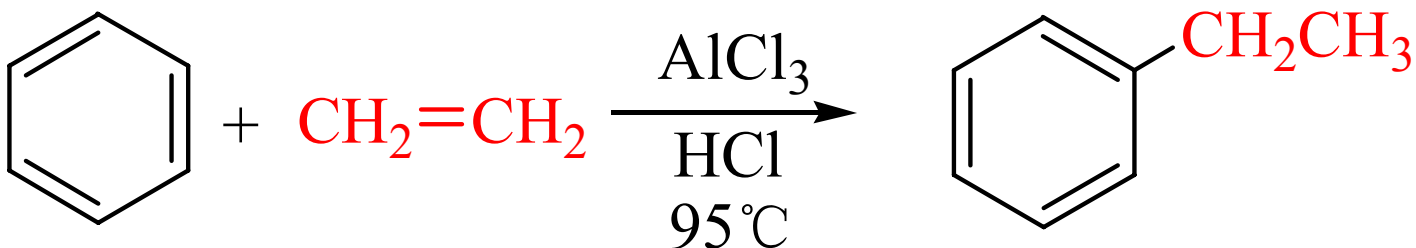
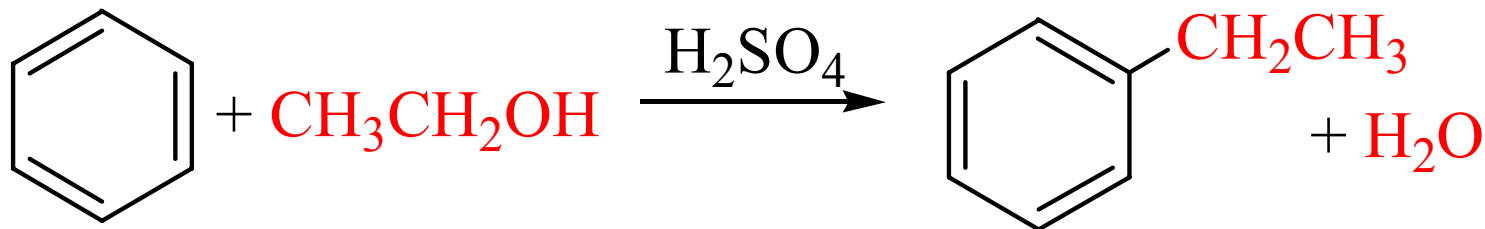
不反应

厦门大学有机化学B

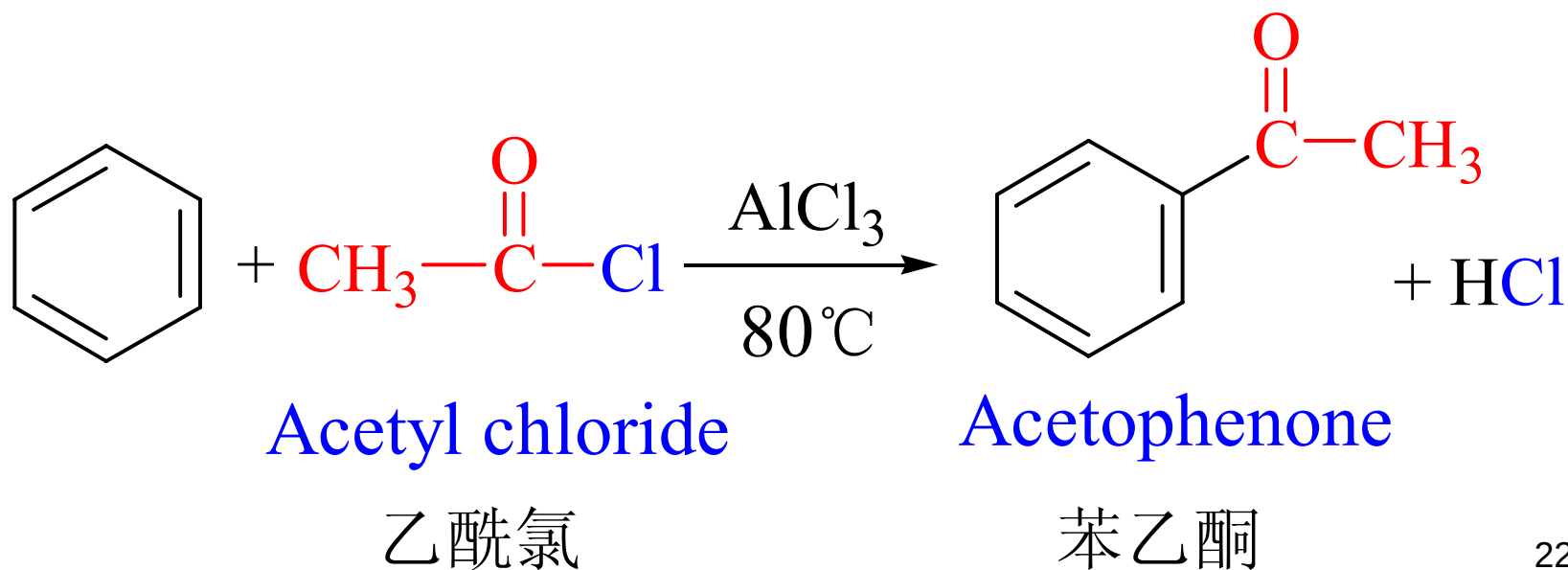
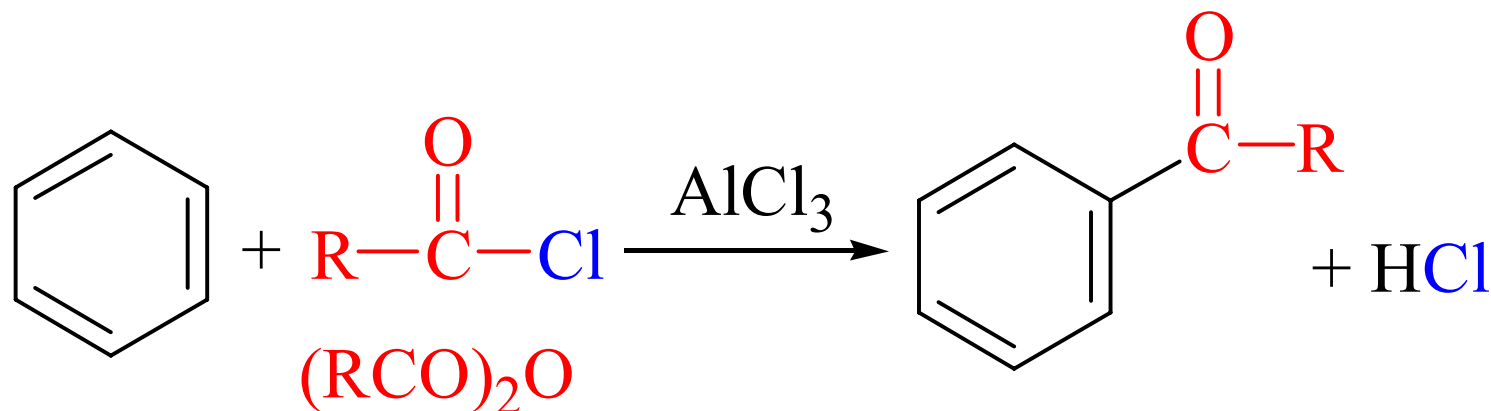
反应机理



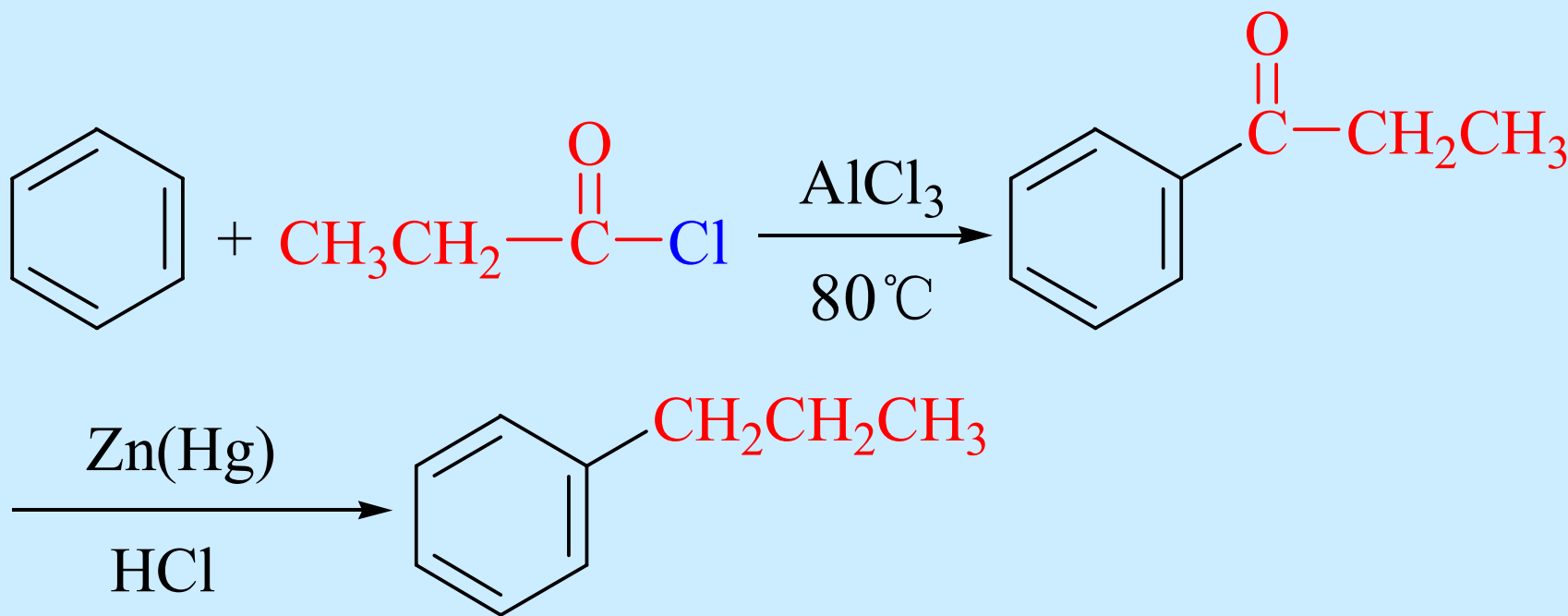
烷基化试剂除卤代烃外，还可以用醇、烯烃



(B) 酰基化(Acylation)

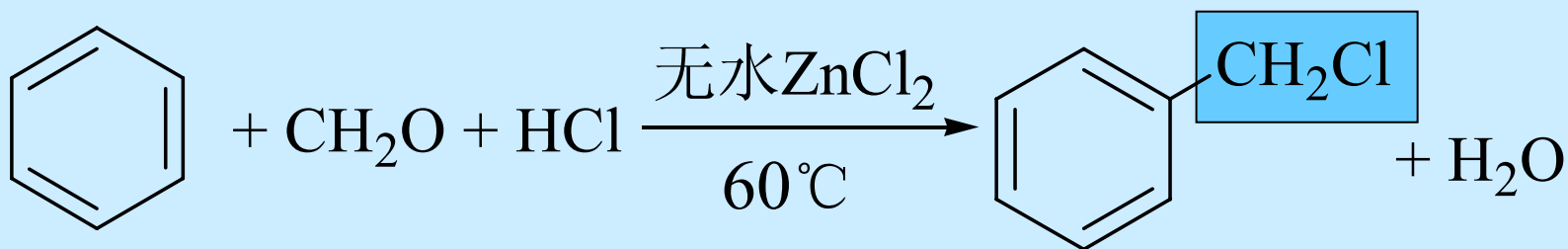


酰基化—还原反应系列

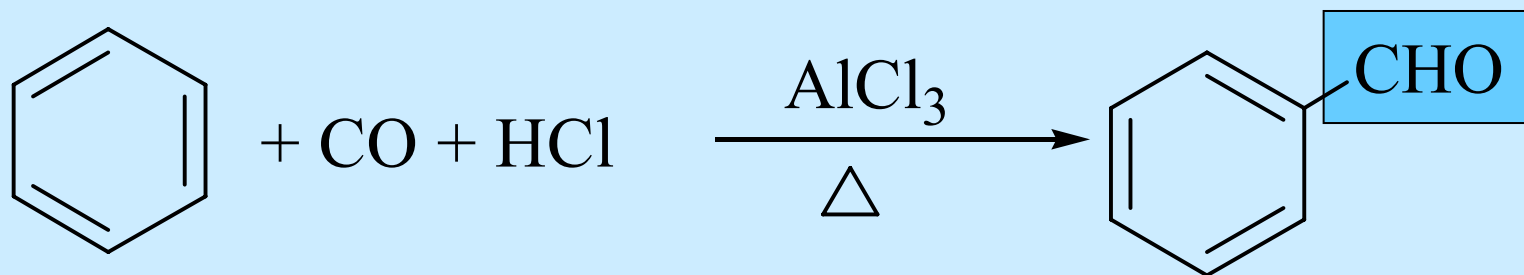


Clemmenson reduction

克莱门森还原



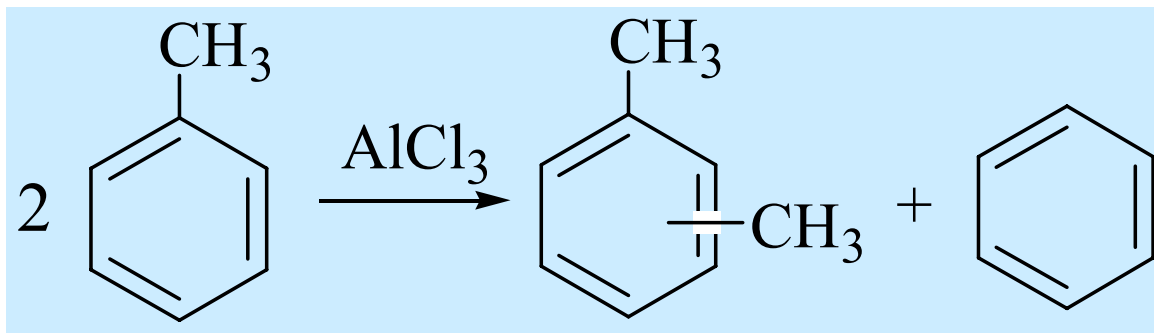
氯甲基化 (Chloromethylation)



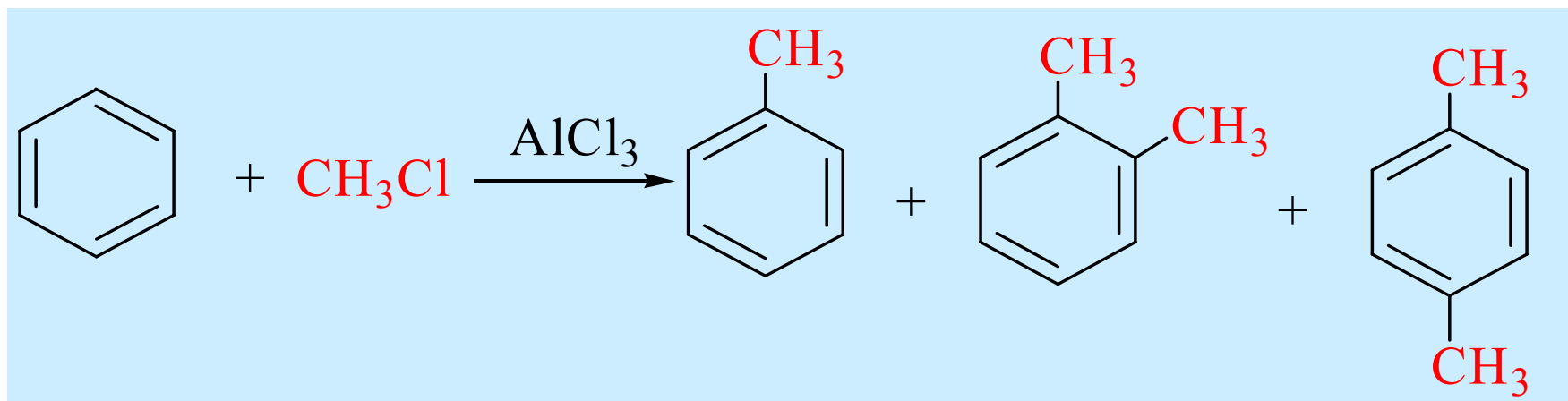
Gettnerman—Koch反应

Friedel—Crafts反应局限性

- 烷基化是可逆反应

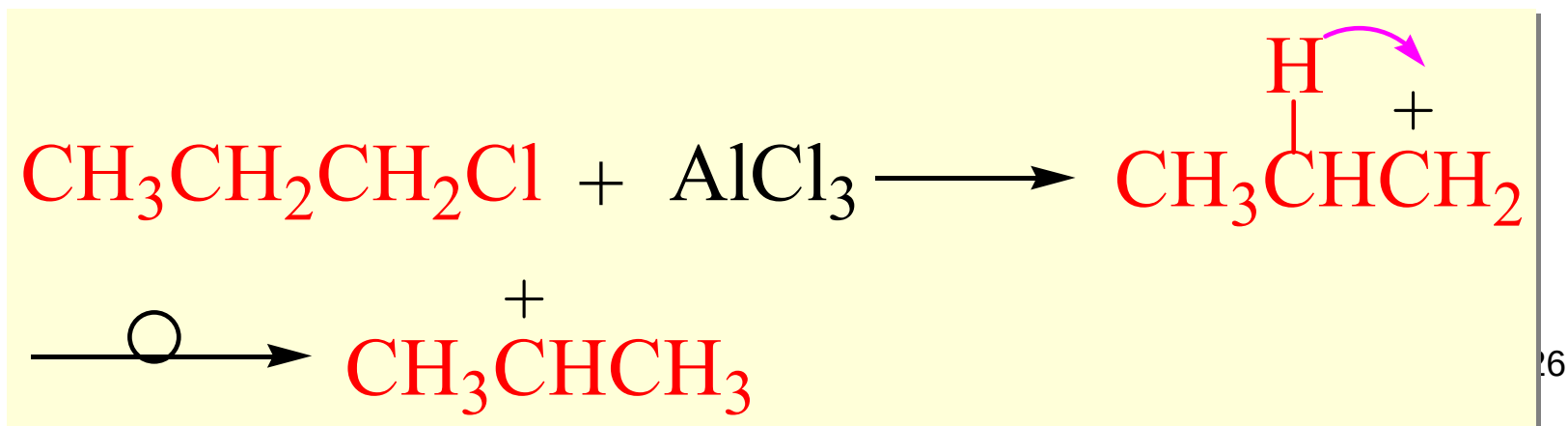
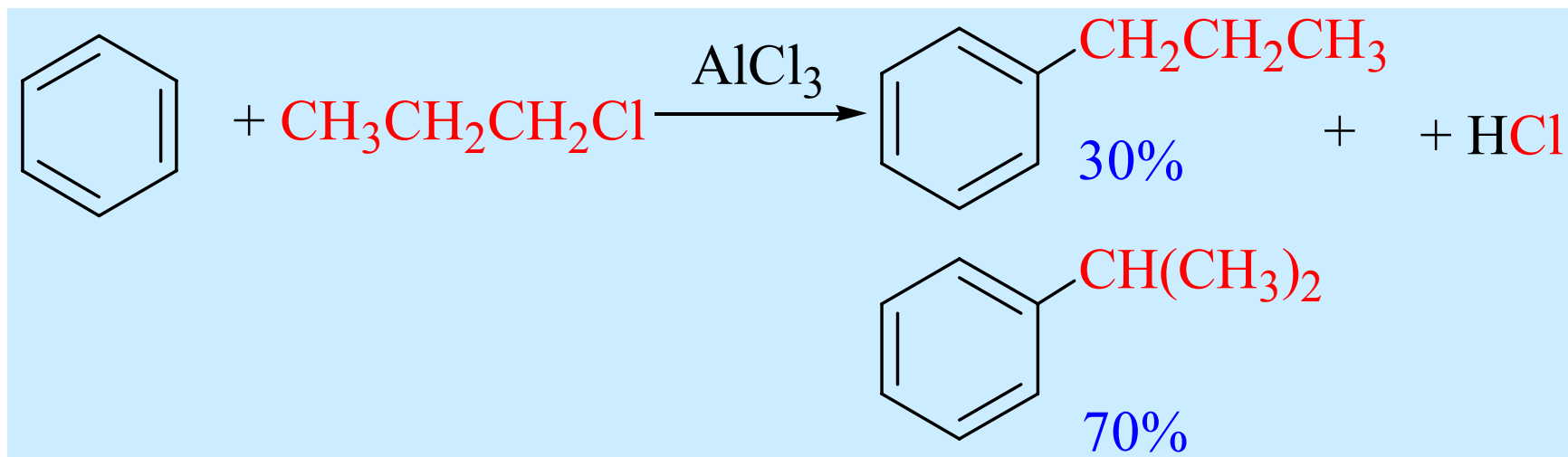


- 易发生多取代现象



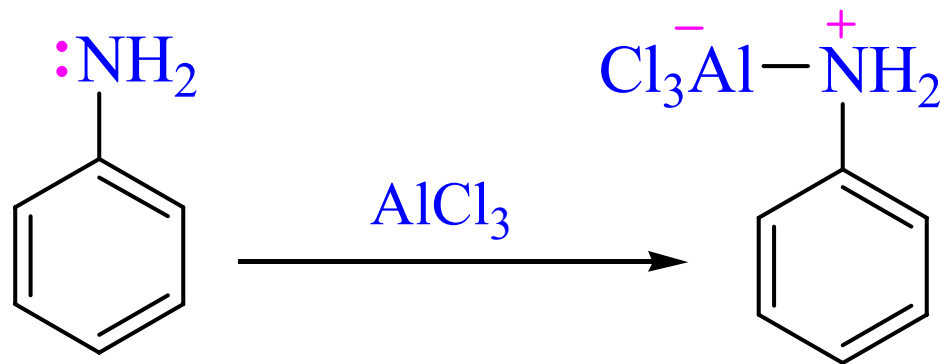
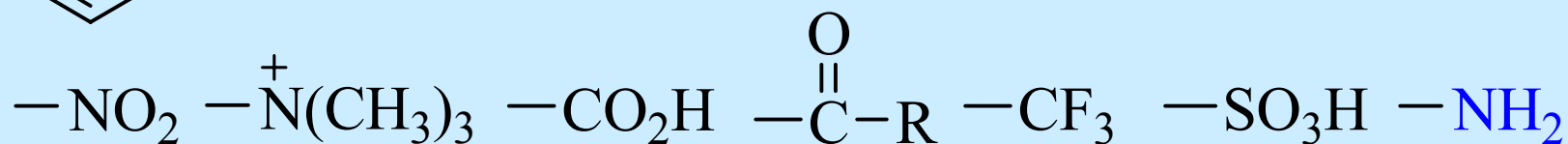
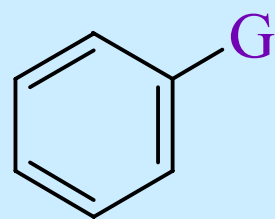
Friedel—Crafts反应局限性

- 当RX超过3个碳原子时，可能发生重排反应



Friedel—Crafts反应局限性

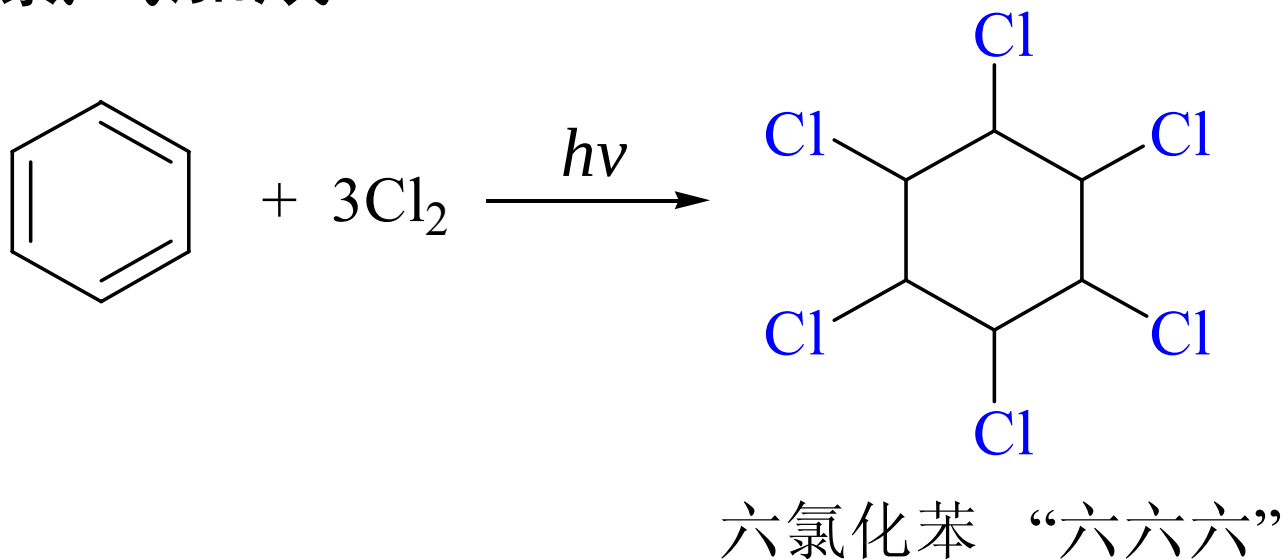
- 当苯环上有强吸电子取代基时，不发生Friedel—Crafts反应



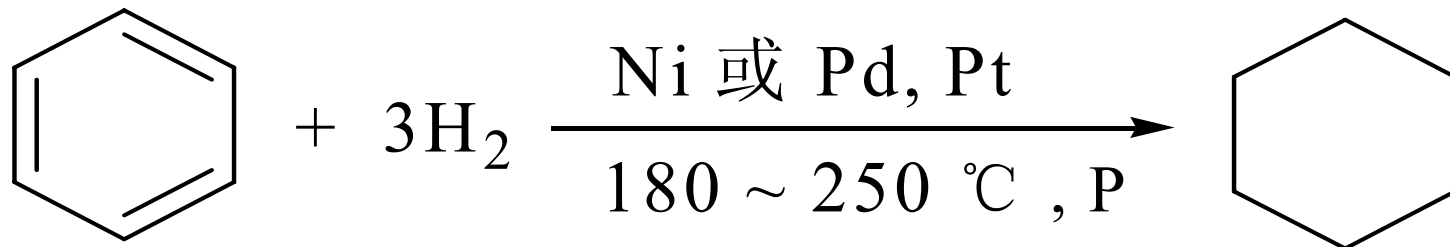
不发生
Friedel-Crafts 反应

6.3.2 加成与还原反应

- 与氯气加成

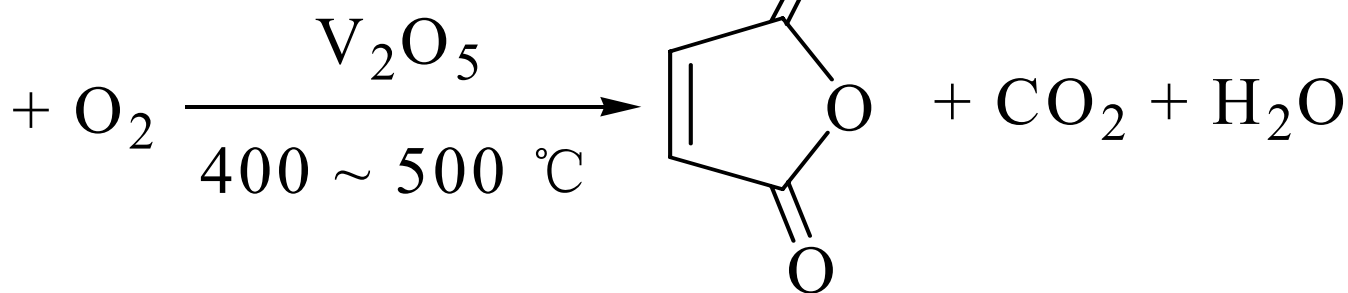
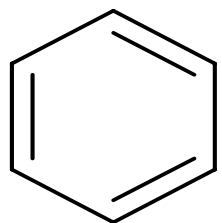
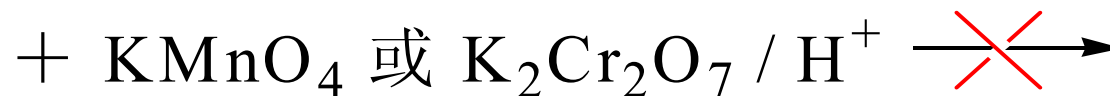
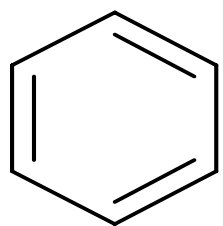


- 加氢



6.3.3 氧化反应

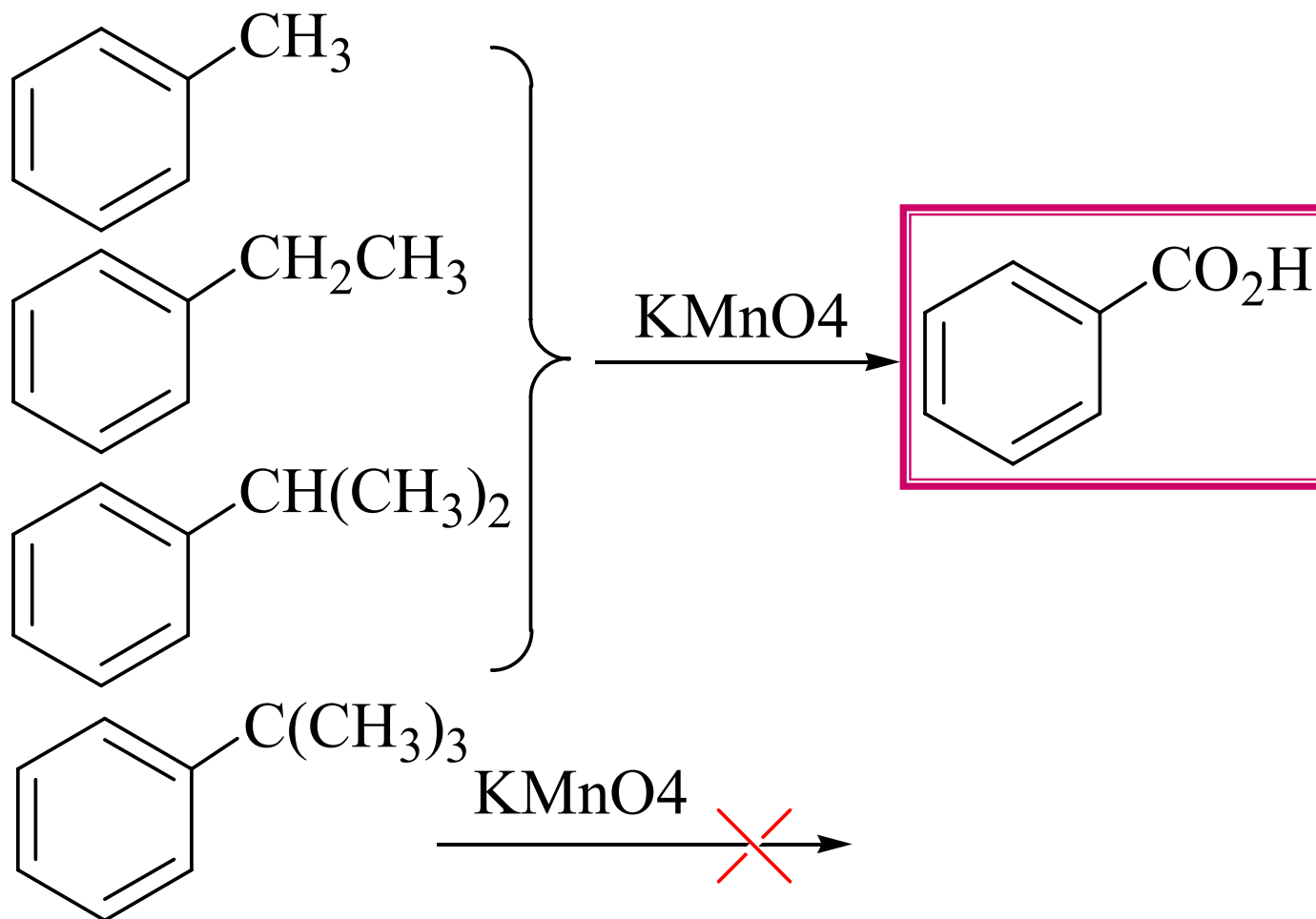
- 苯环的氧化

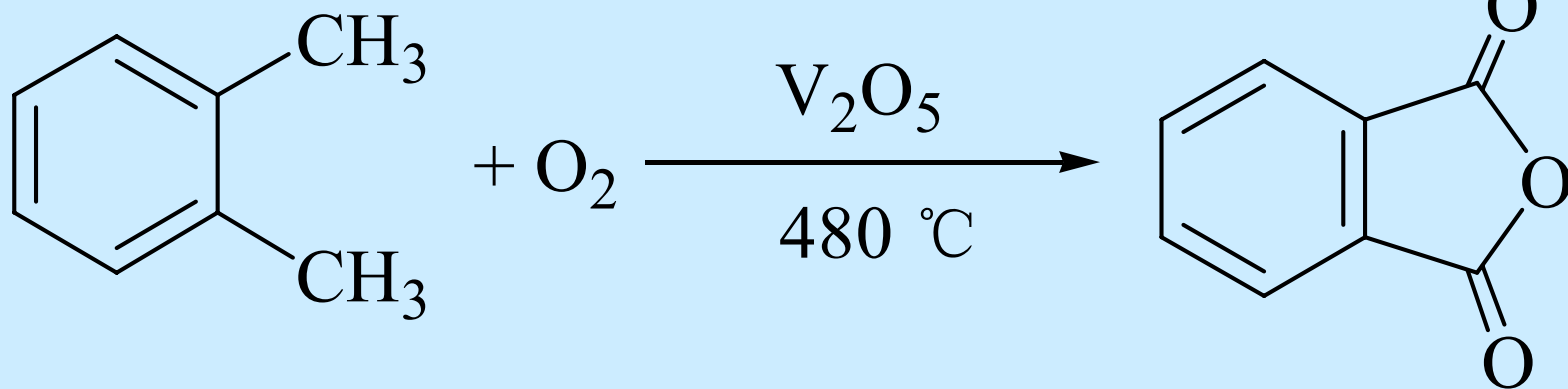
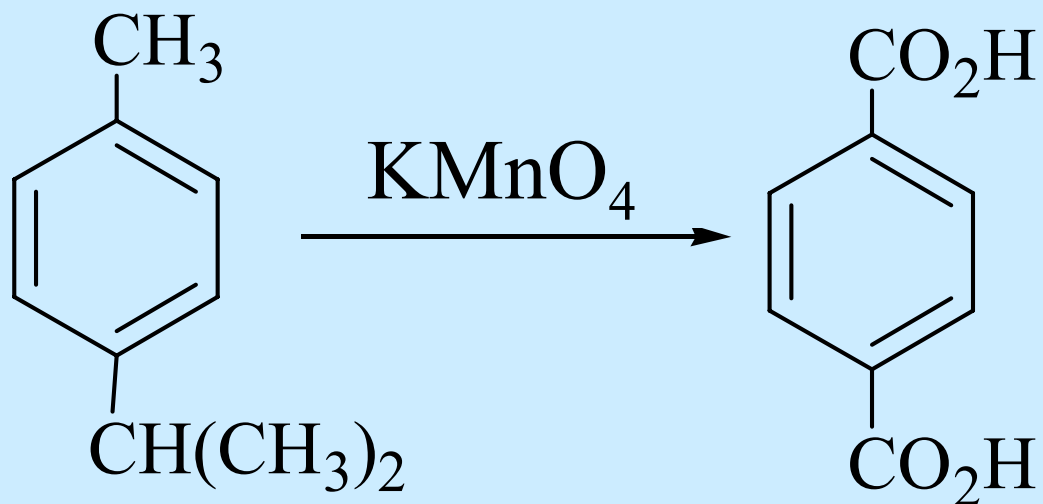


顺丁烯二酸酐
Maleic anhydride

6.3.3 氧化反应

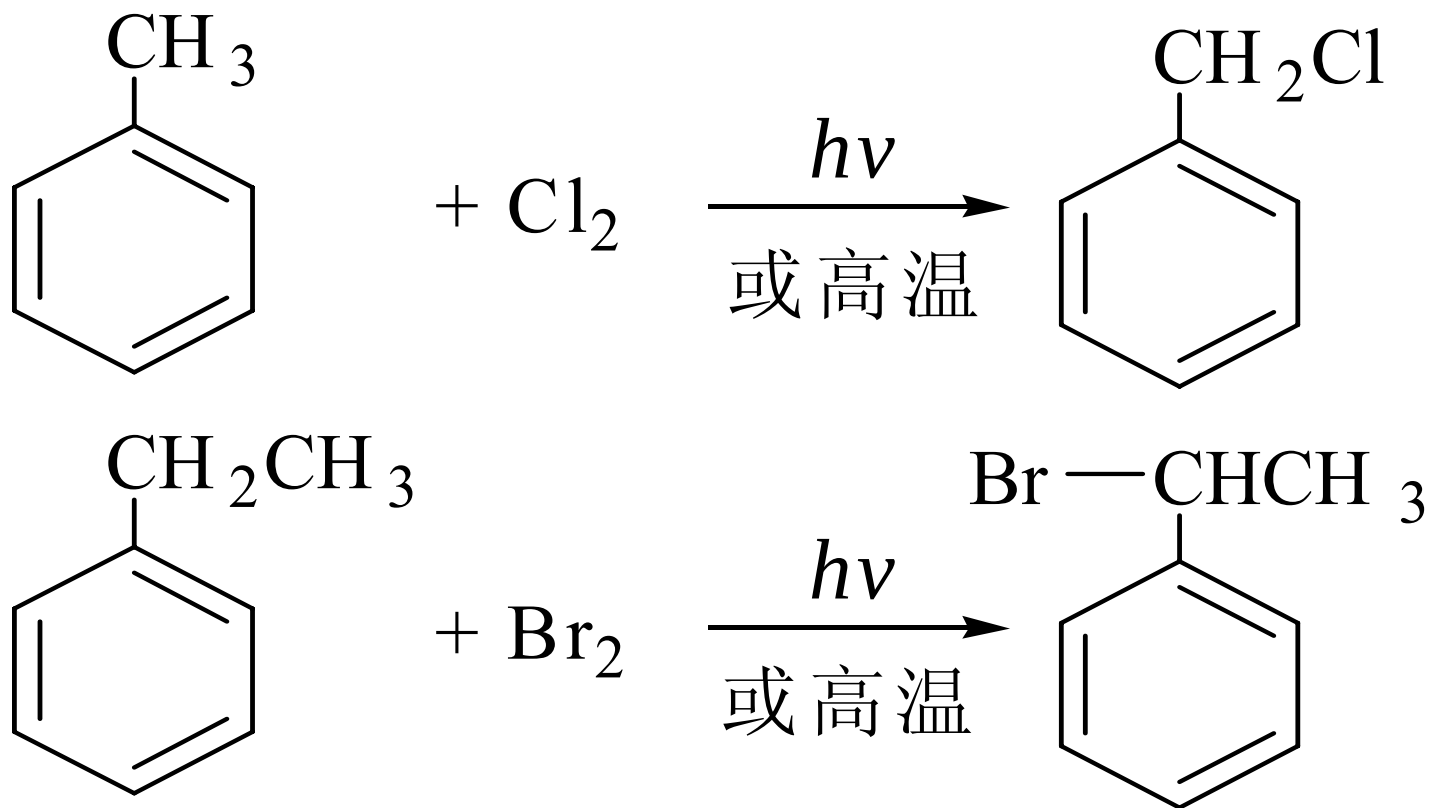
- 侧链氧化





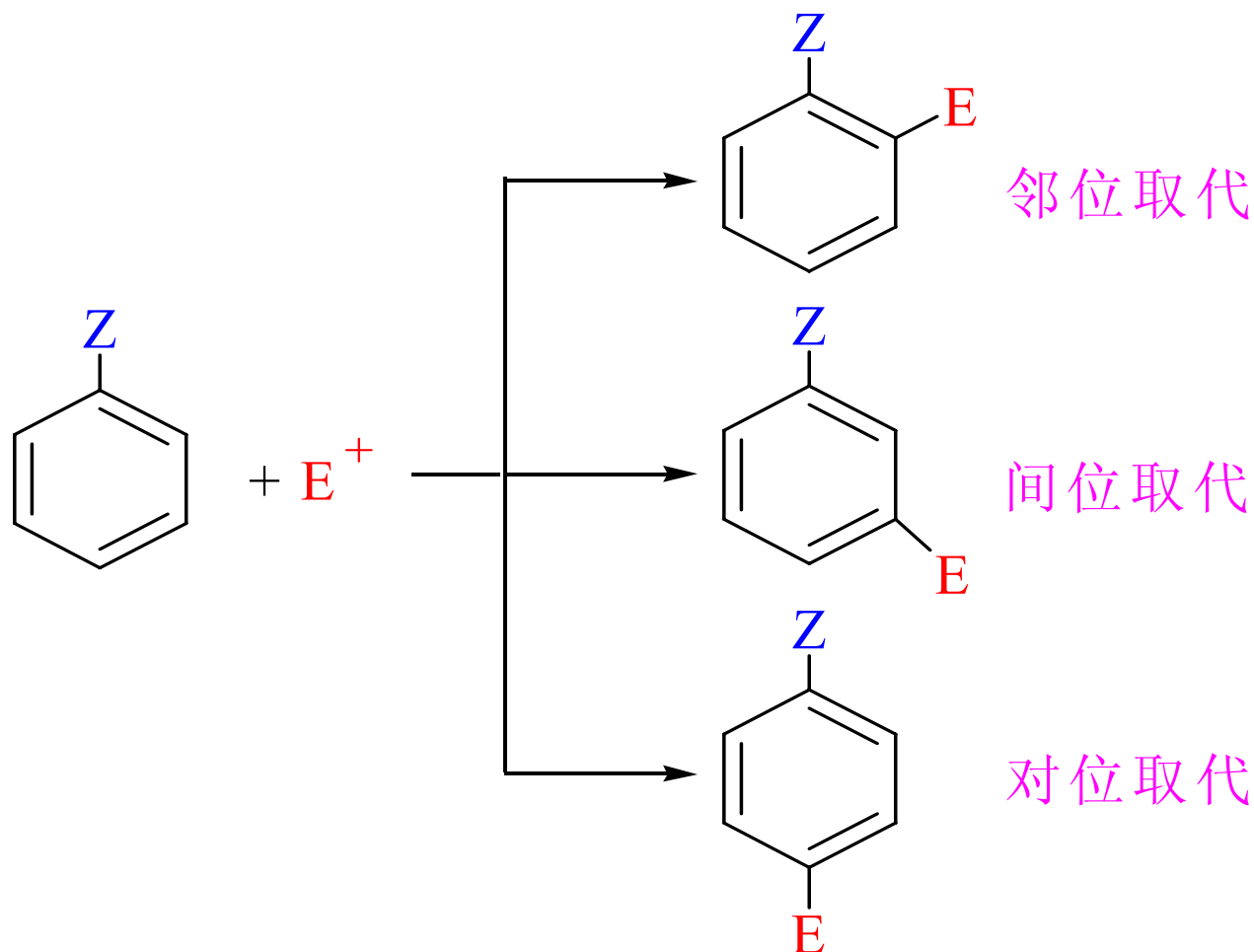
邻苯二甲酸酐

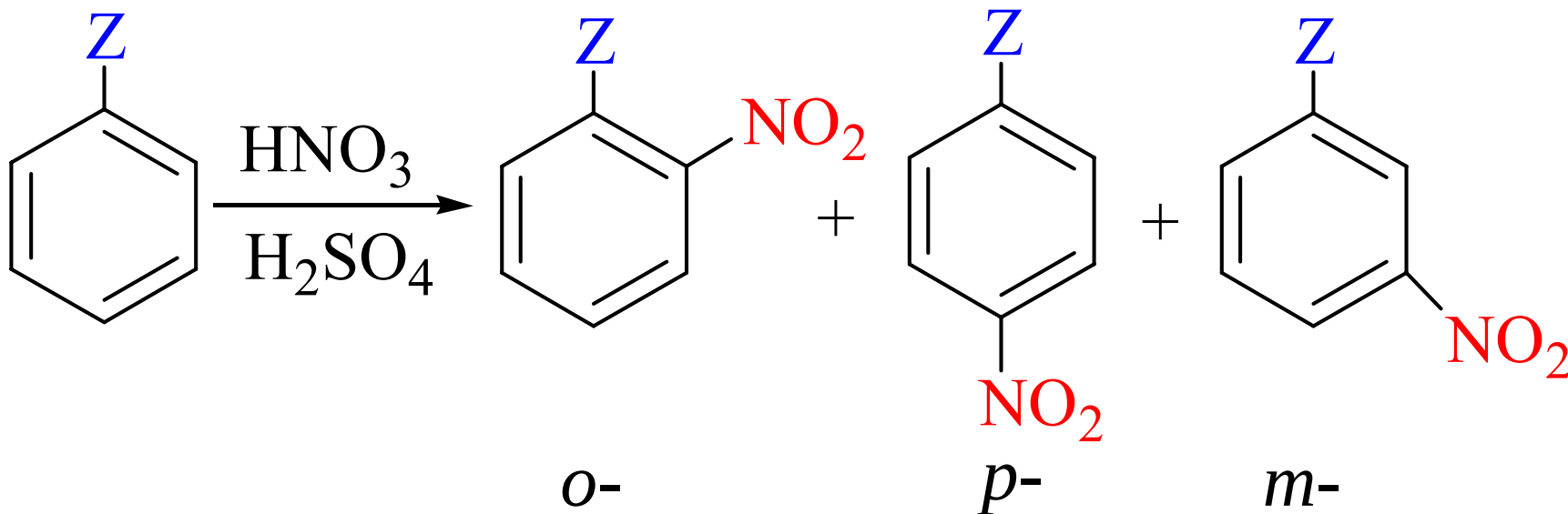
6.3.4 α -卤代反应



6.4 苯环上亲电取代的定位规律

6.4.1 定位效应(Orientation)



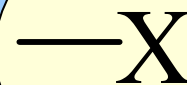
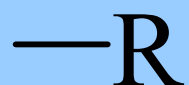
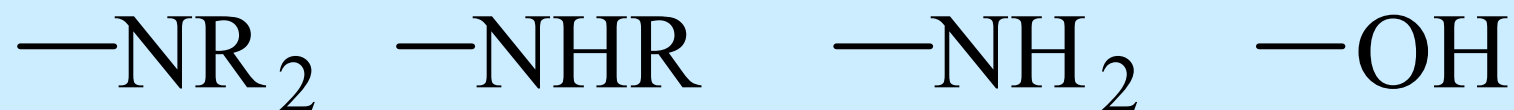


Z=	OH	Me	Cl	NO₂	CN
o(%)	40	58	30	6	17
p(%)	60	38	69	<1	2
m(%)	<1	4	1	93	81

定位效应：取代基进入的位置受原有取代基的影响

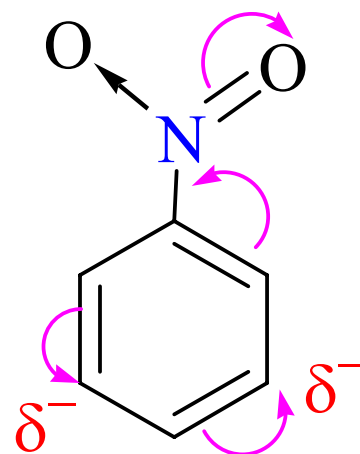
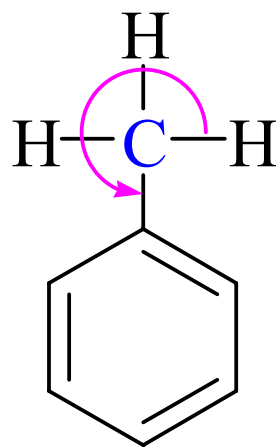
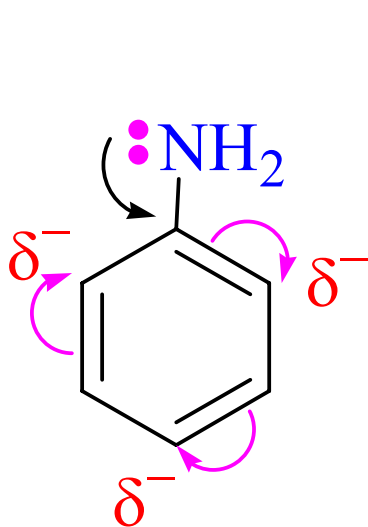
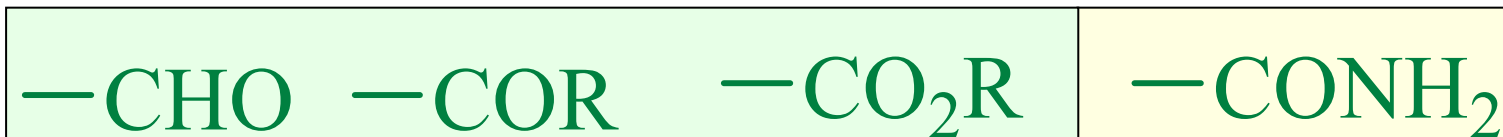
6.4.2 两类定位基团

1. 第一类定位基(邻对位定位基)



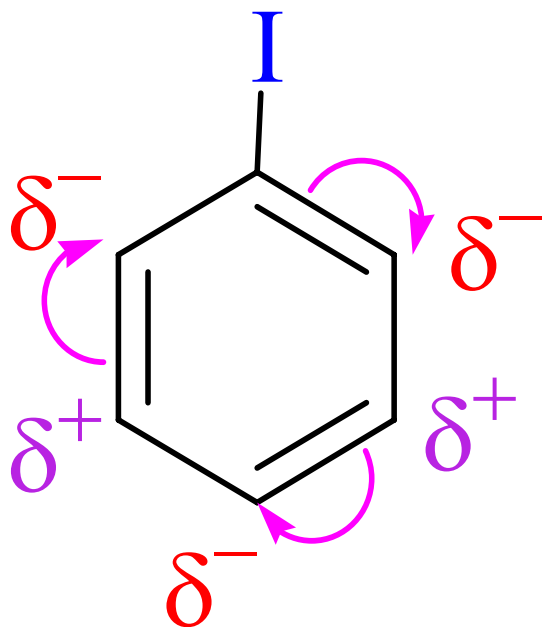
6.4.2 两类定位基团

2. 第二类定位基(间位定位基)

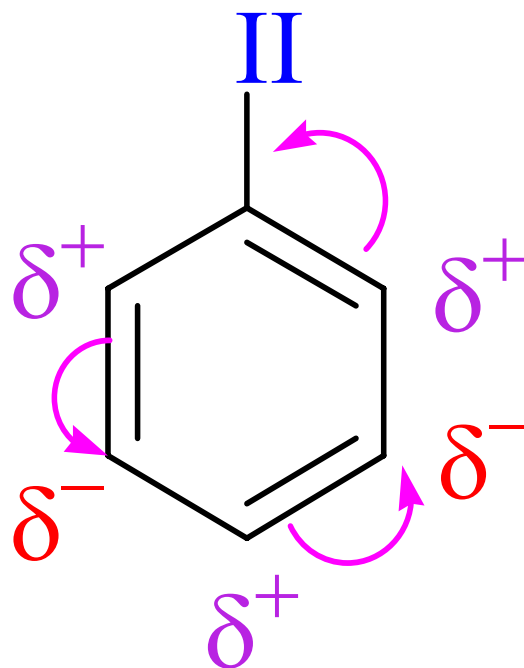


6.4.3 定位规律的理论解释

1. 苯环上的电子分布情况



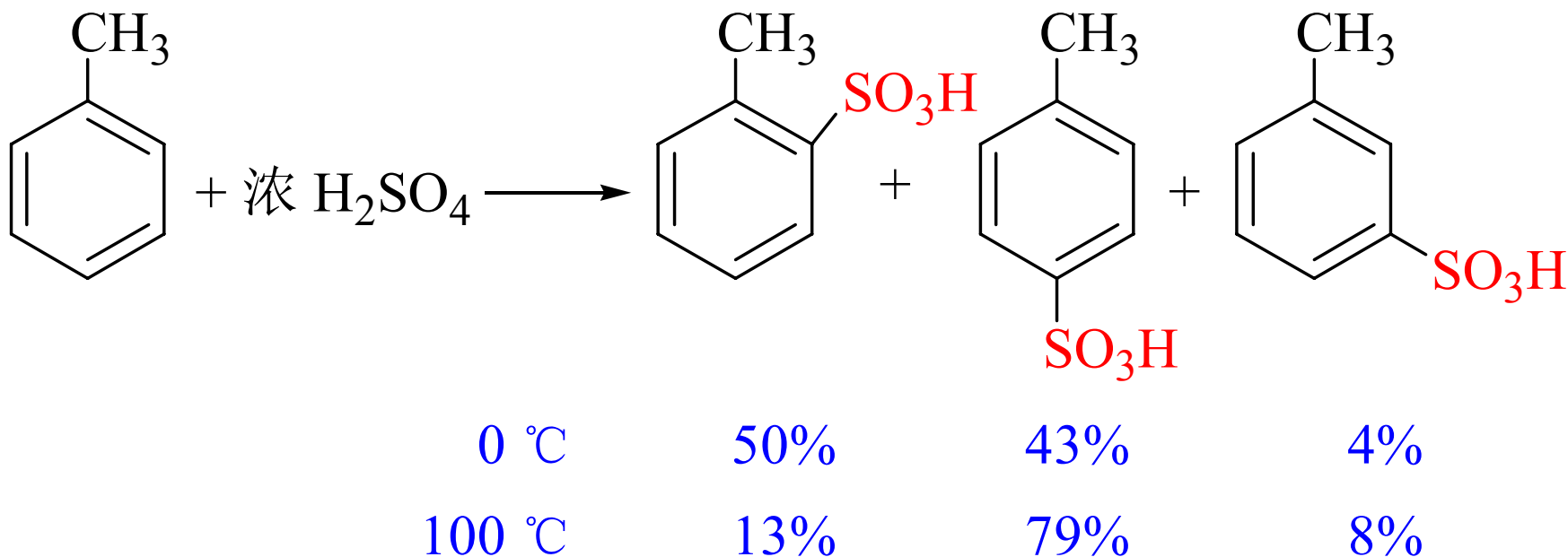
I 类定位基



II 类定位基

6.4.4 影响定位效应的因素(了解)

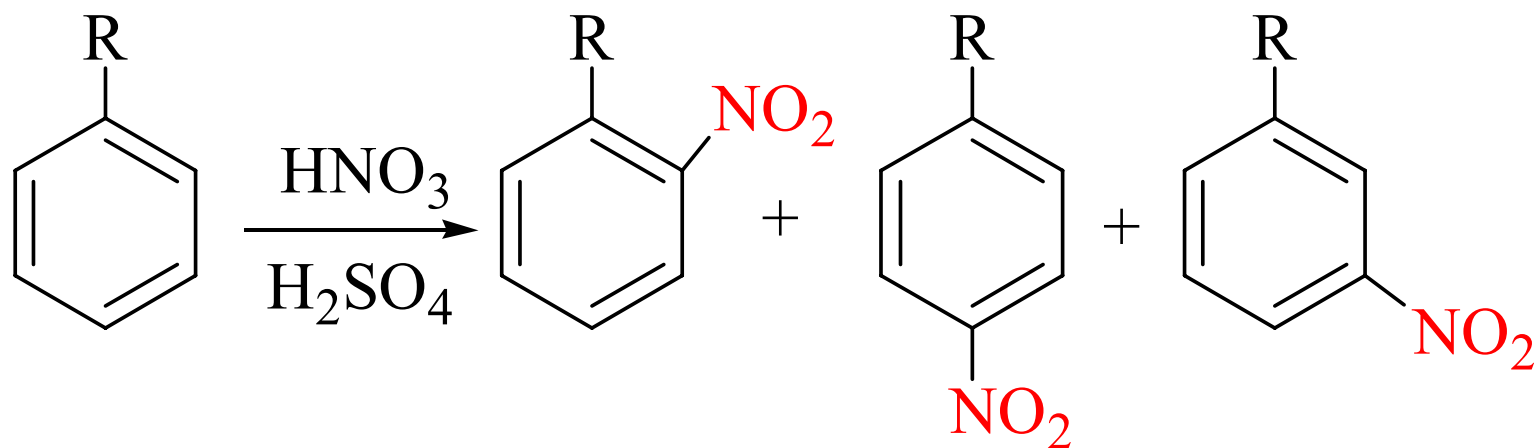
1. 温度的影响



6.4.4 影响定位效应的因素(了解)

2. 空间效应的影响

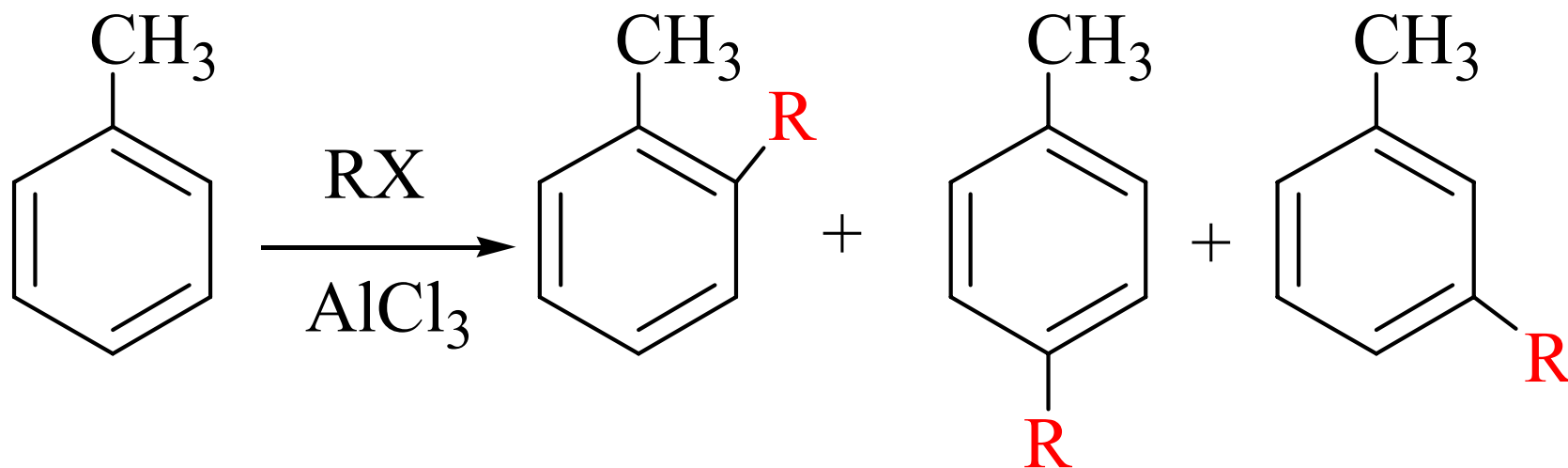
取代基体积增大，邻位产物减少，对位产物增多



$\text{R} = \text{CH}_3$	58.4%	37.2%	4.4%
CH_2CH_3	45.0%	48.5%	6.5%
$\text{CH}(\text{CH}_3)_2$	30.0%	62.5%	7.7%
$\text{C}(\text{CH}_3)_3$	15.8%	72.7%	11.5%

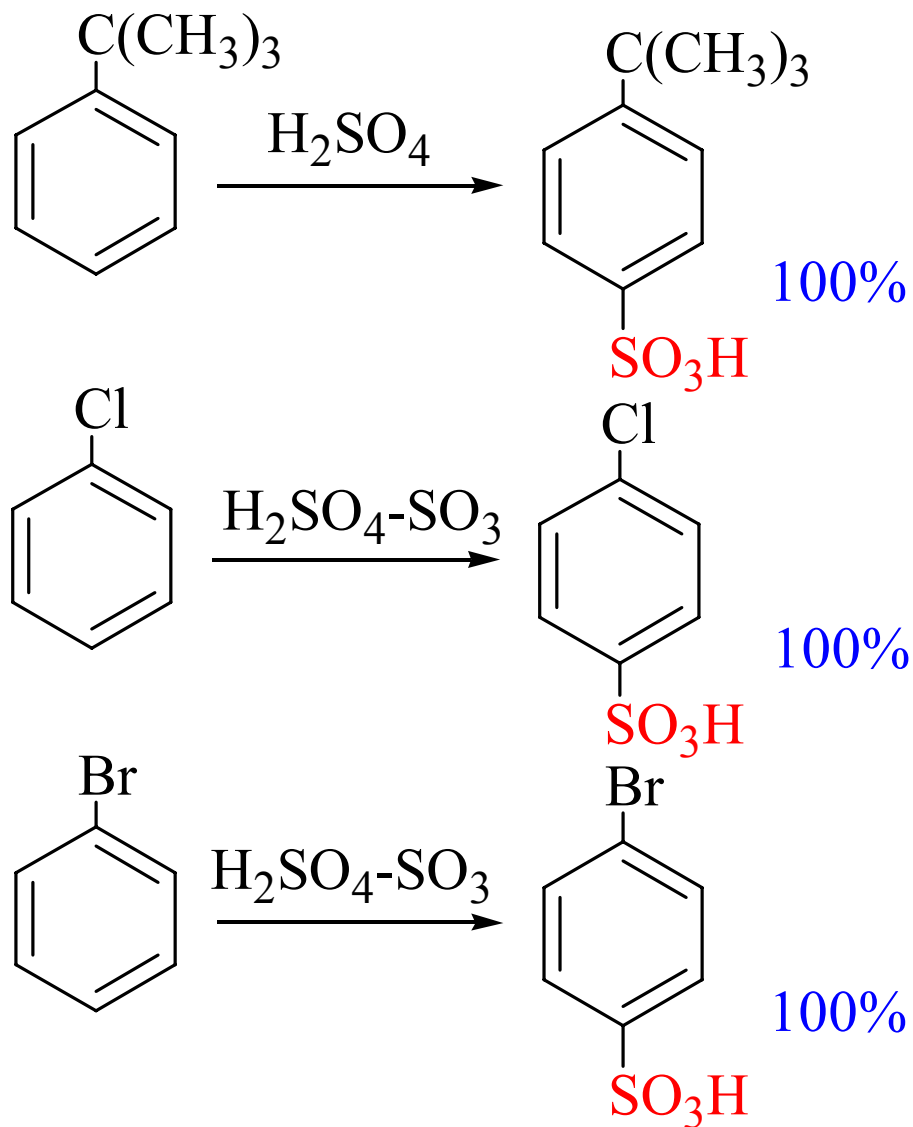
6.4.4 影响定位效应的因素(了解)

3.试剂的体积



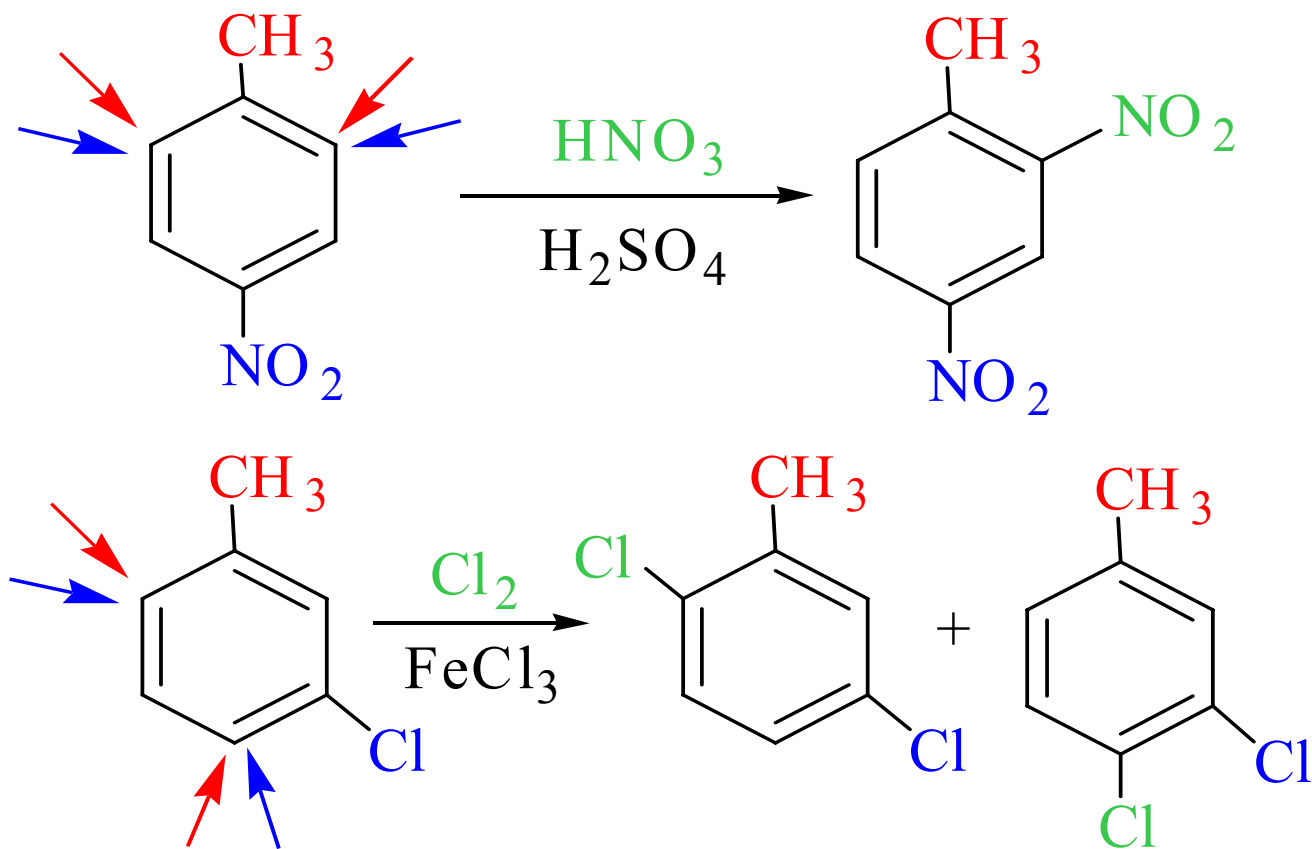
$\text{R} = \text{CH}_3$	53.8%	28.8%	17.3%
CH_2CH_3	45%	25%	30%
$\text{CH}(\text{CH}_3)_2$	37%	32.7%	29.8%
$\text{C}(\text{CH}_3)_3$	0%	93%	7%

取代基与试剂体积都大时，邻位产物极少



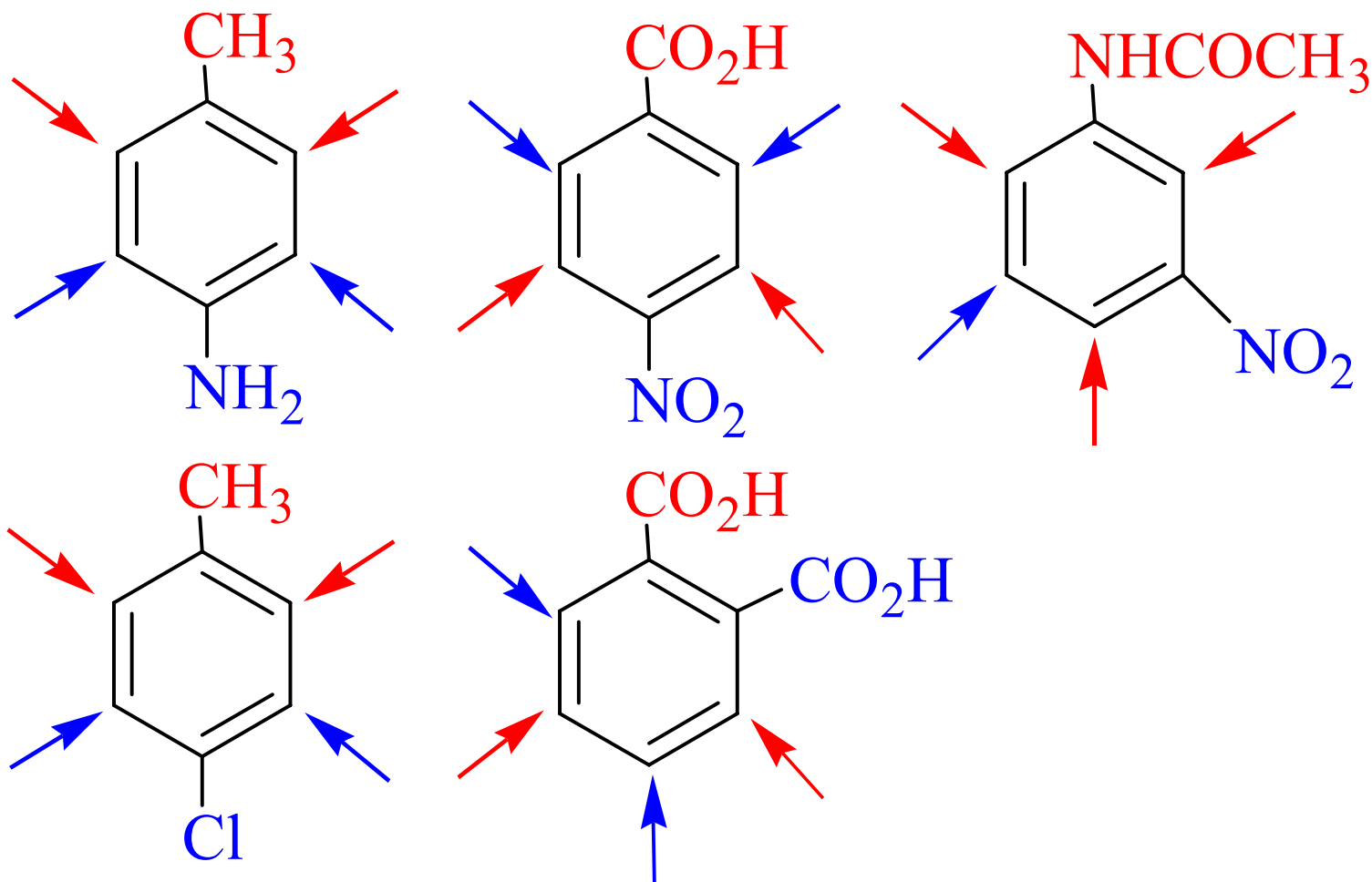
6.4.5 二元取代苯的定位

1. 两个取代基定位效应一致

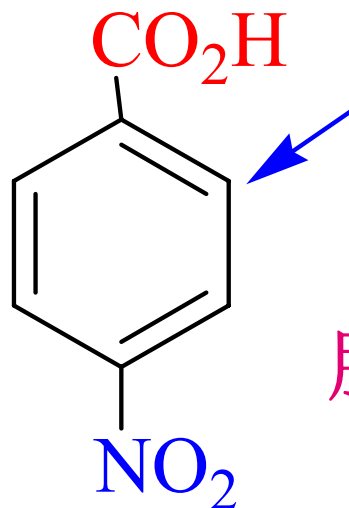
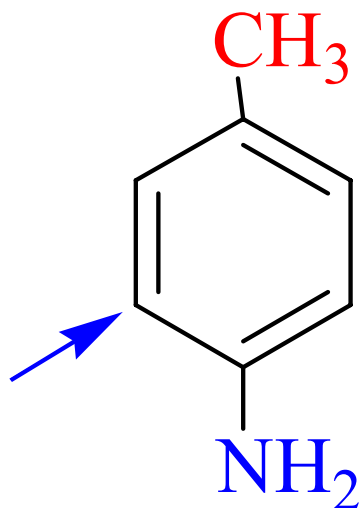


6.4.5 二元取代苯的定位

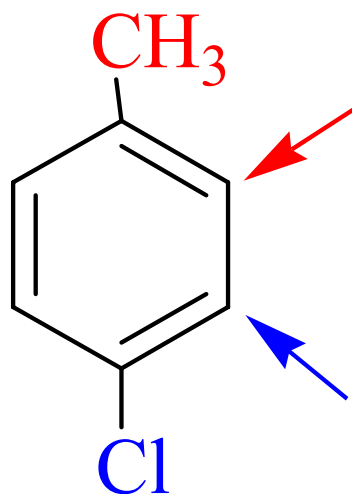
2. 两个取代基定位效应不一致



两个取代基同类

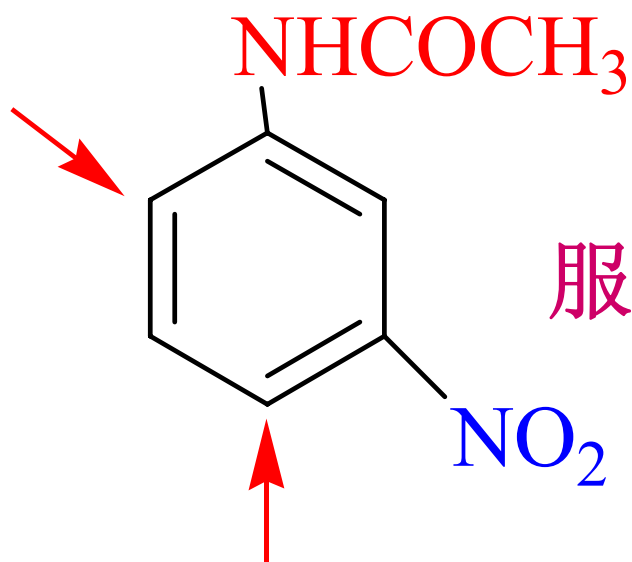


服从较强取代基的定位



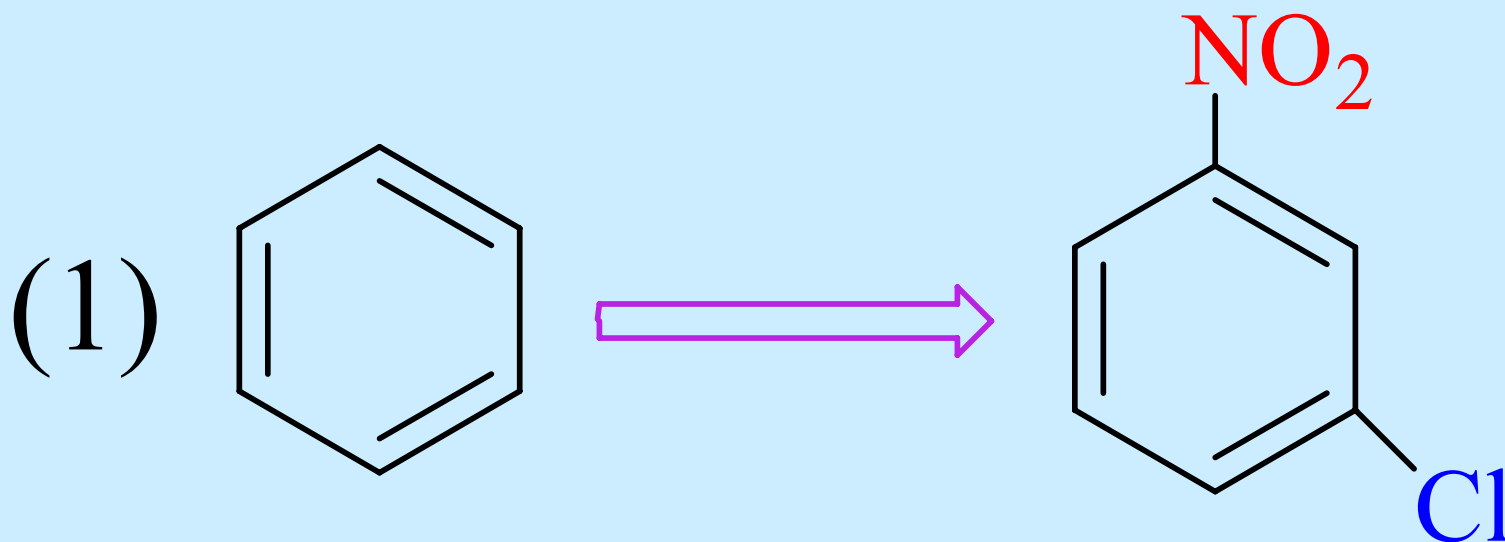
定位能力差不多时
两者都有

两个取代基不同类



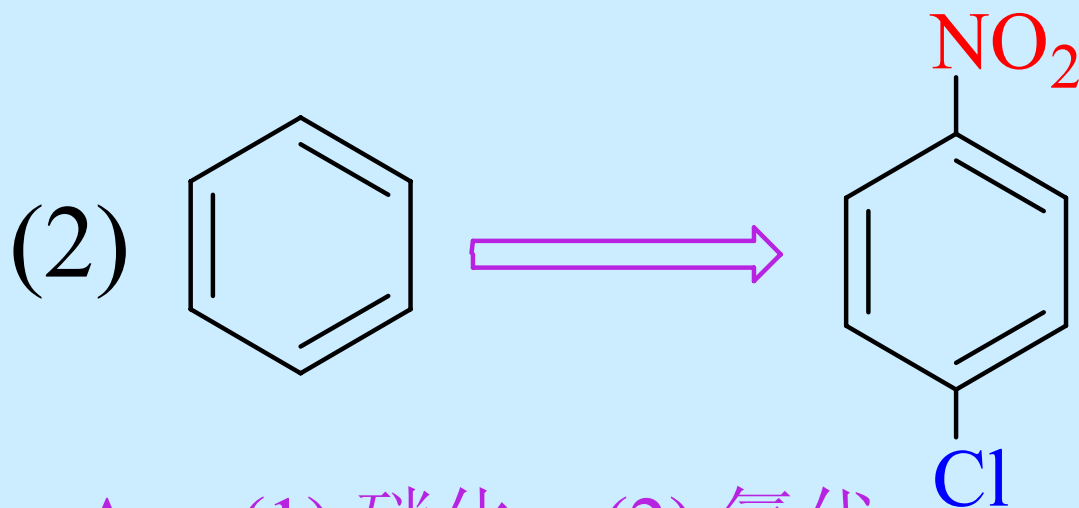
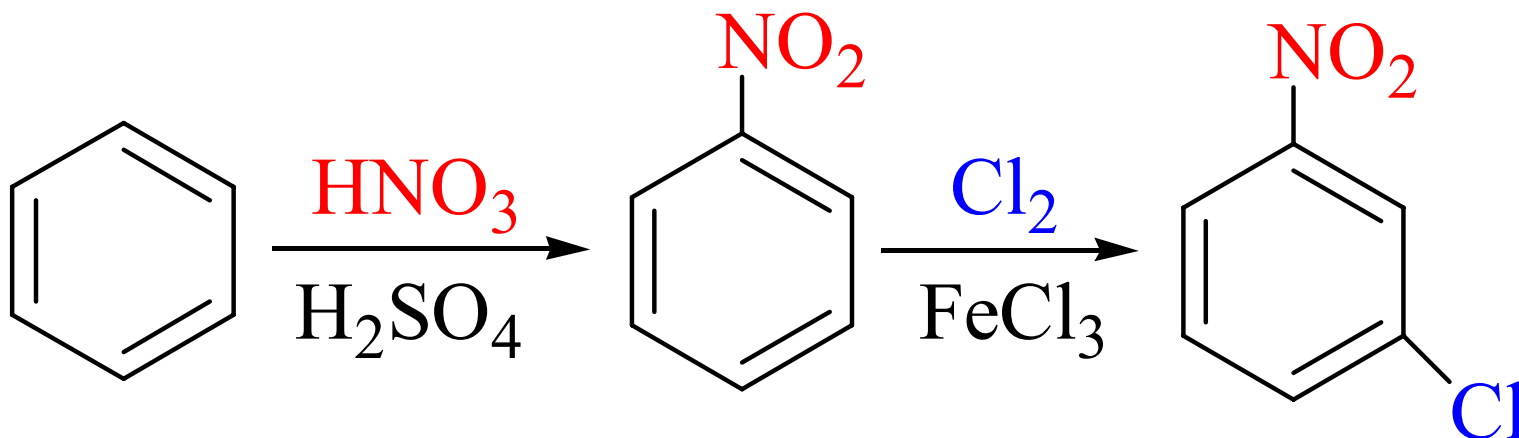
服从一类定位基的定位

6.4.6 定位规律的应用



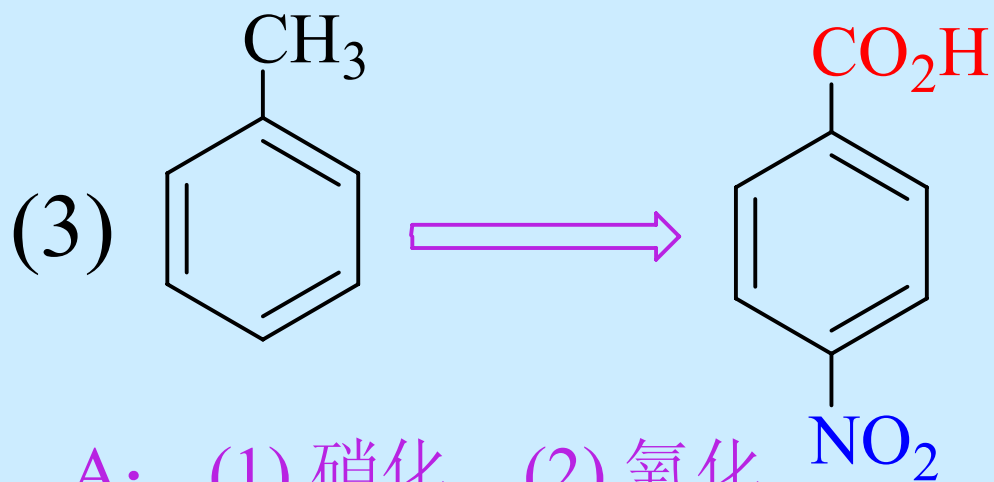
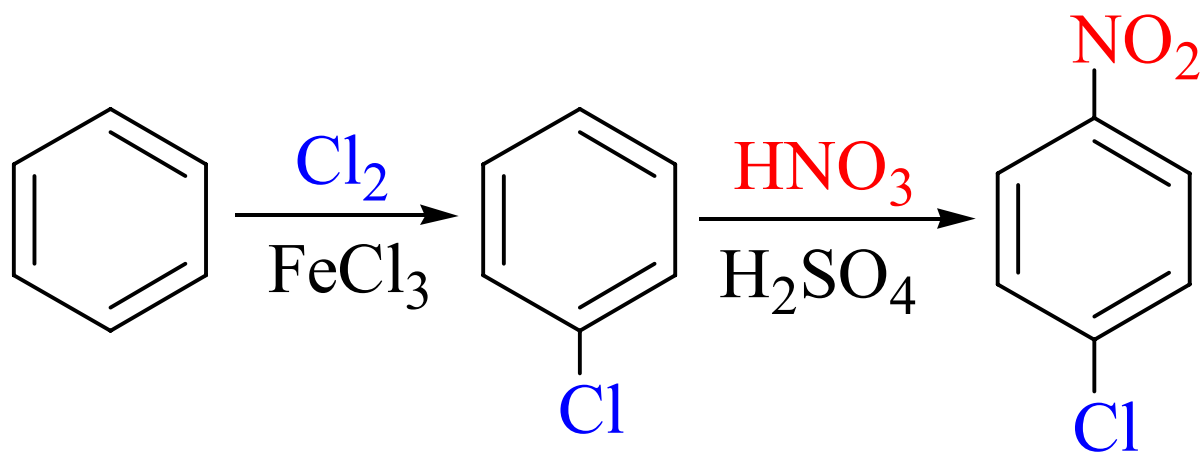
A: (1) 硝化 (2) 氯代

B: (1) 氯代 (2) 硝化



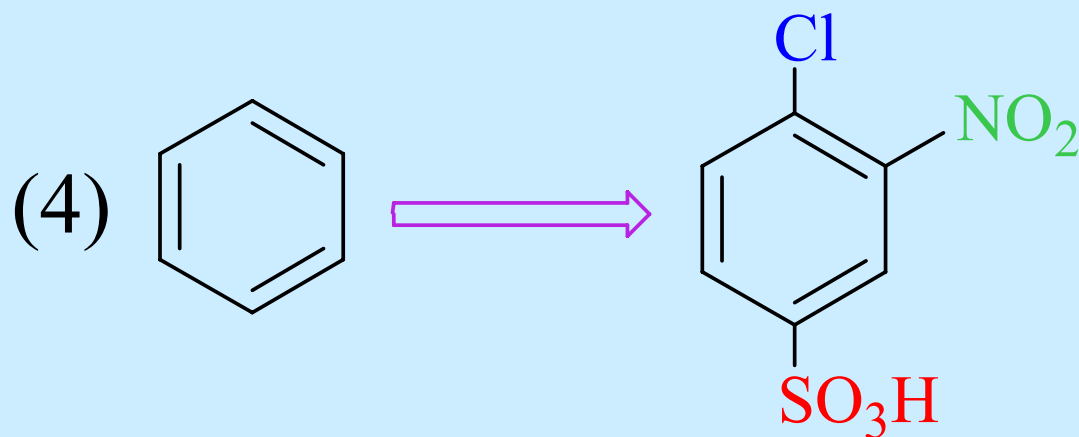
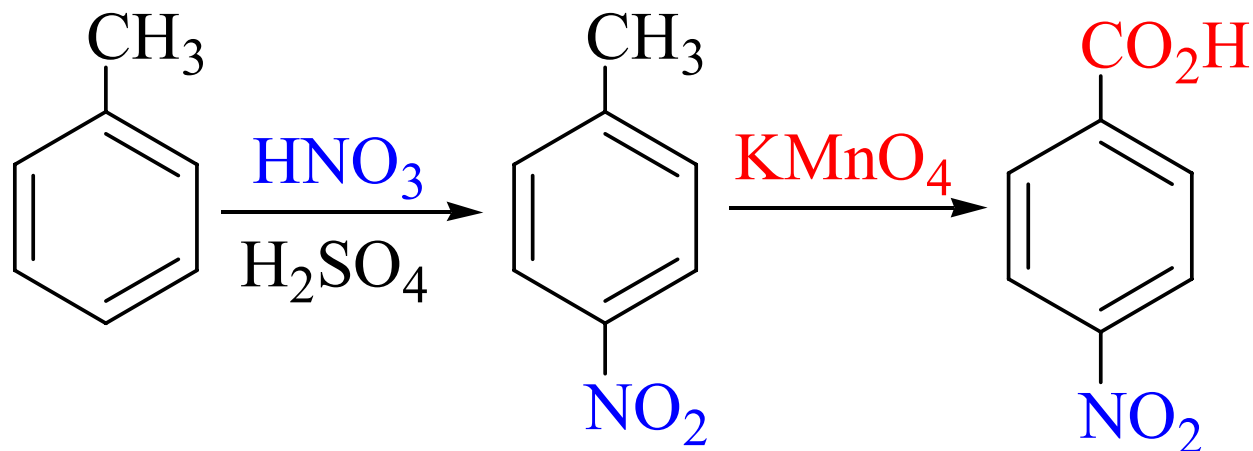
A: (1) 硝化 (2) 氯代

B: (1) 氯代 (2) 硝化



A: (1) 硝化 (2) 氧化

B: (1) 氧化 (2) 硝化

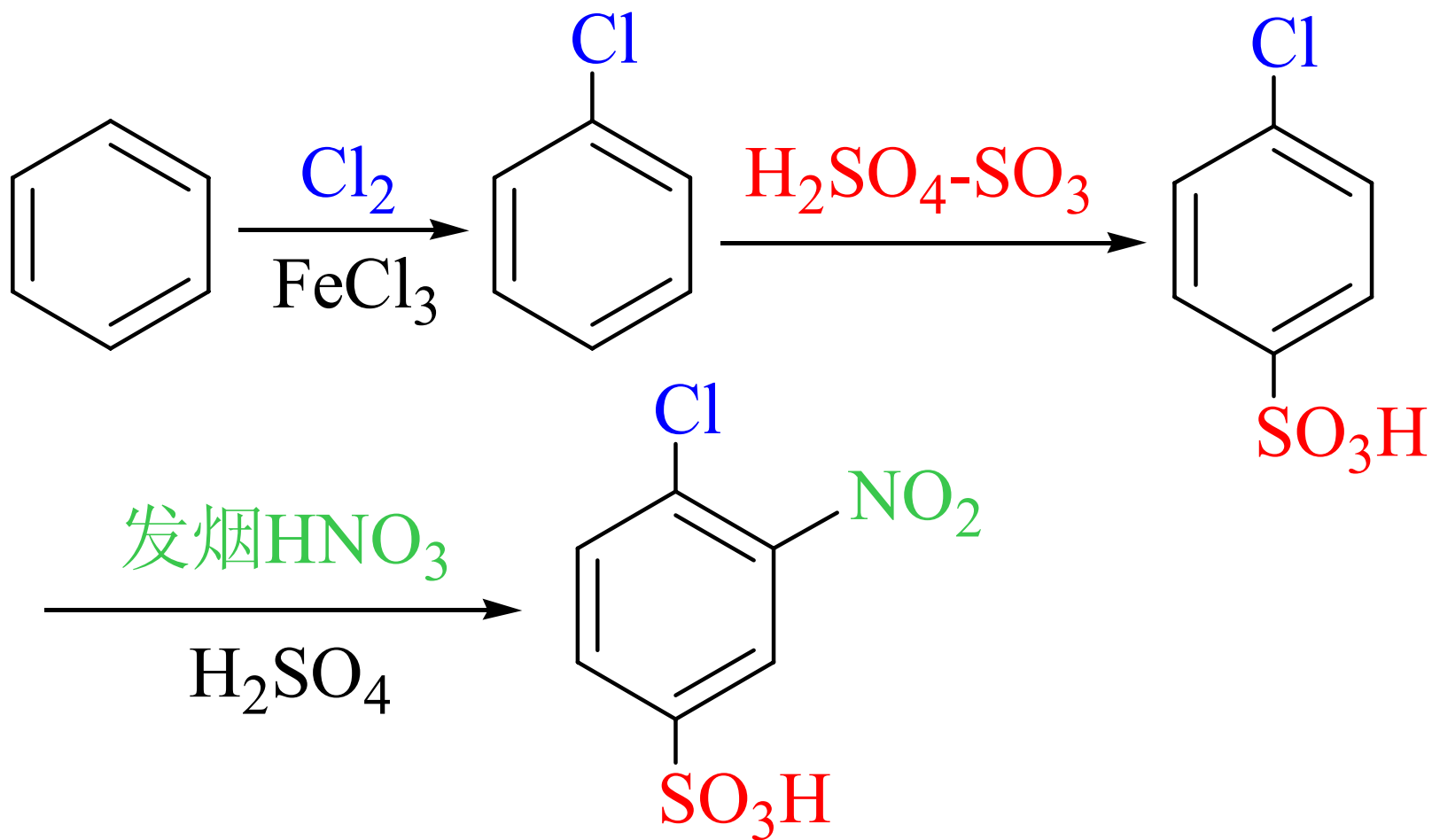


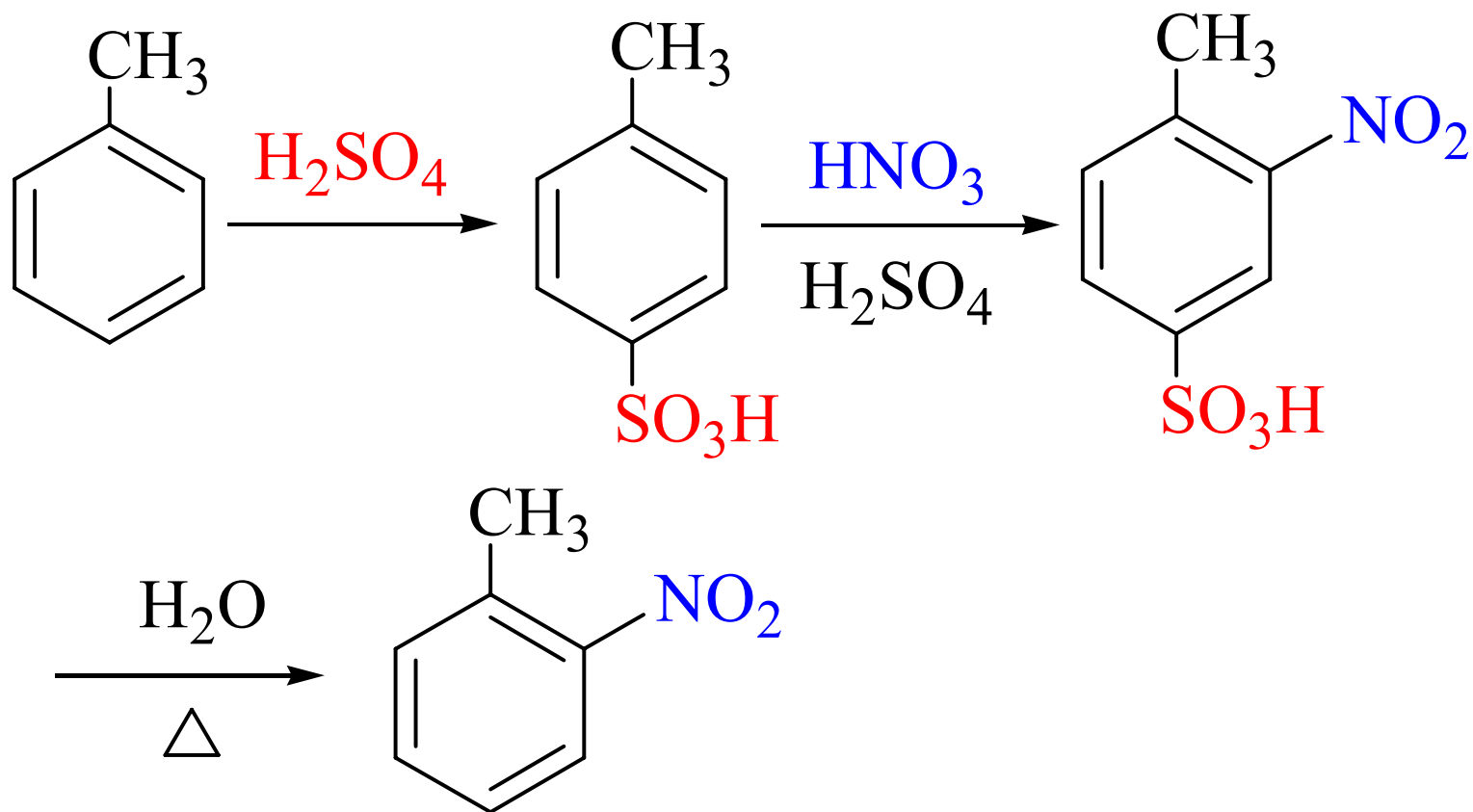
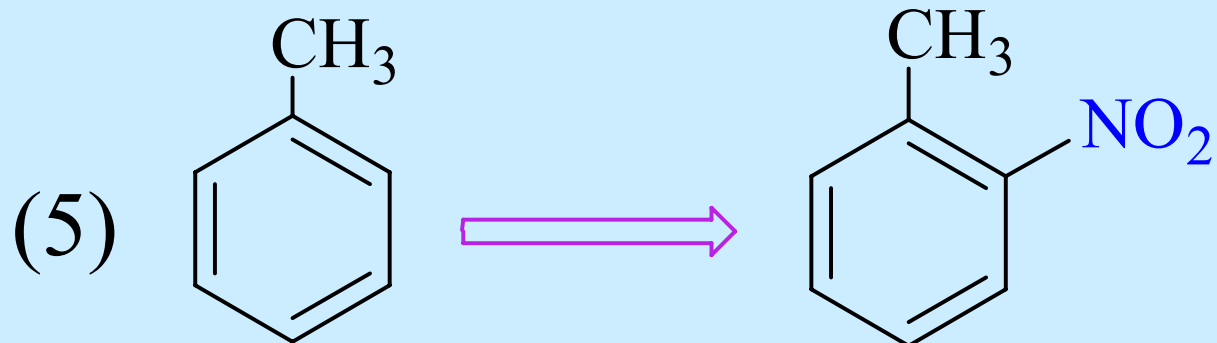
A: (1) 硝化 (2) 氯代 (3) 磺化

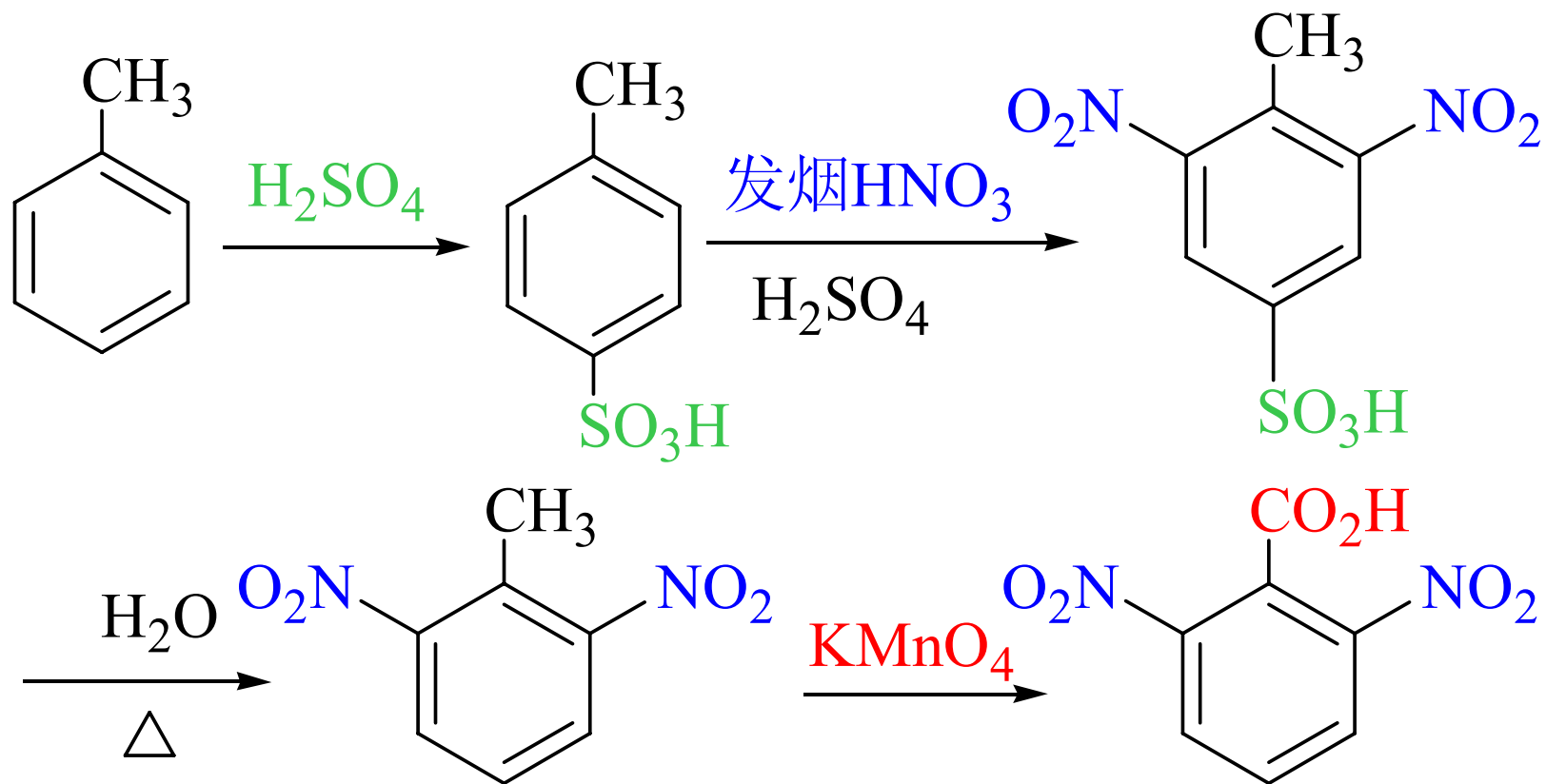
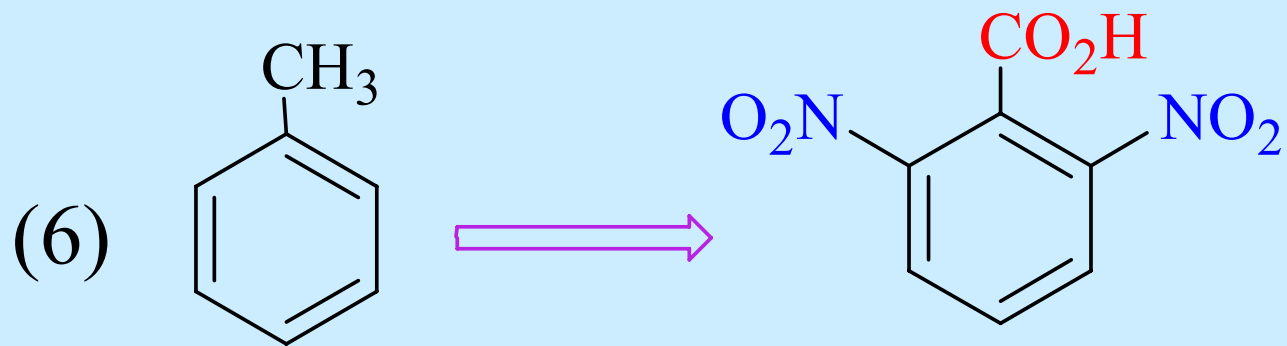
B: (1) 氯代 (2) 硝化 (3) 磺化

C: (1) 硝化 (2) 磺化 (3) 氯代

D: (1) 氯代 (2) 磺化 (3) 硝化

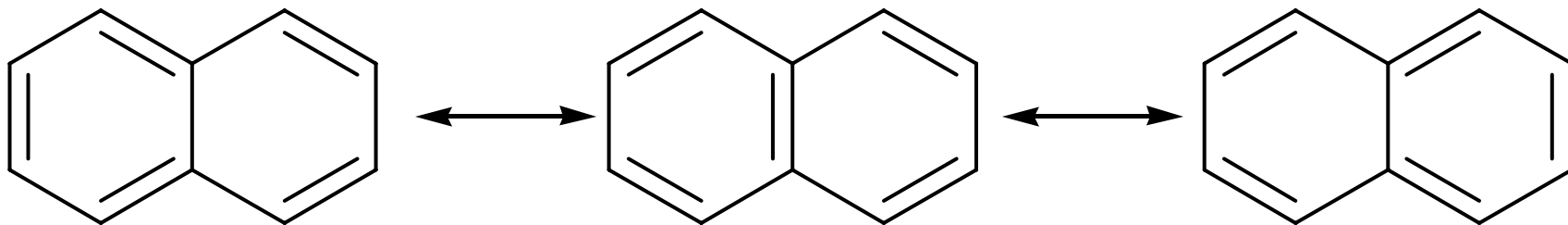
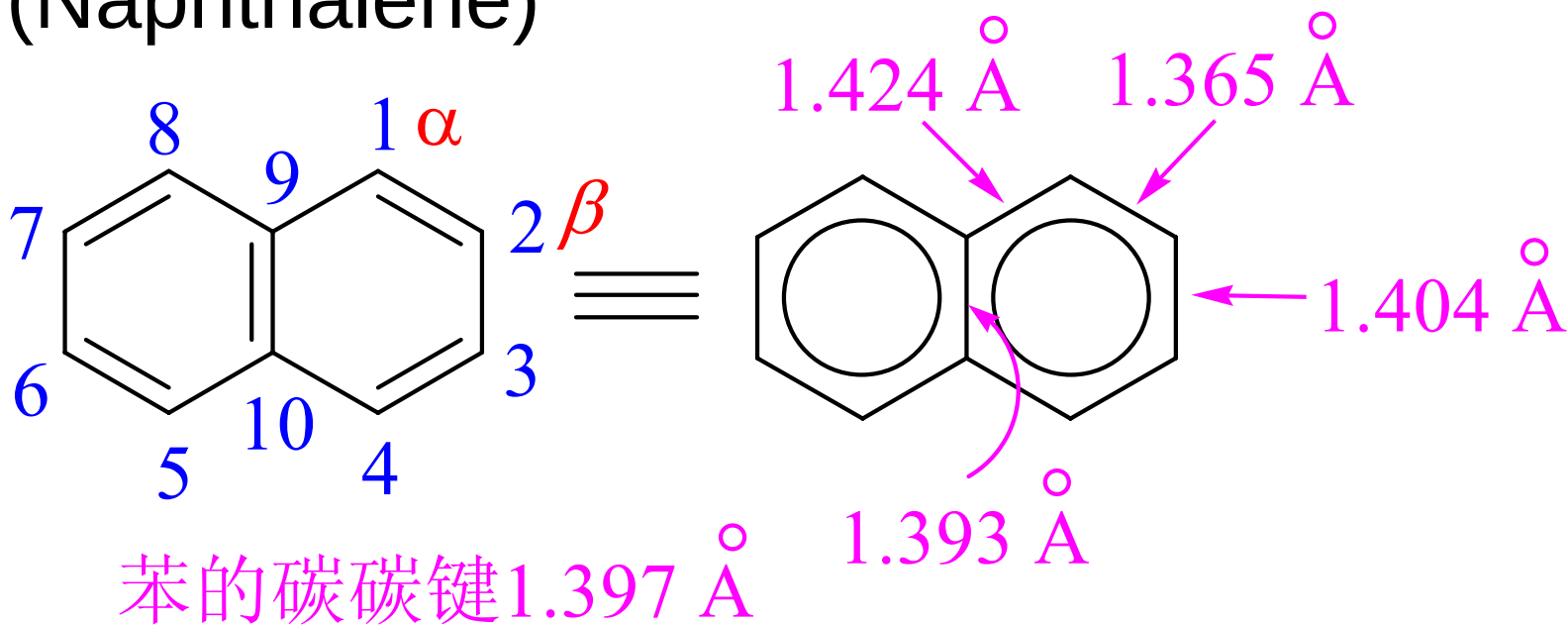






6.5 稠环芳烃

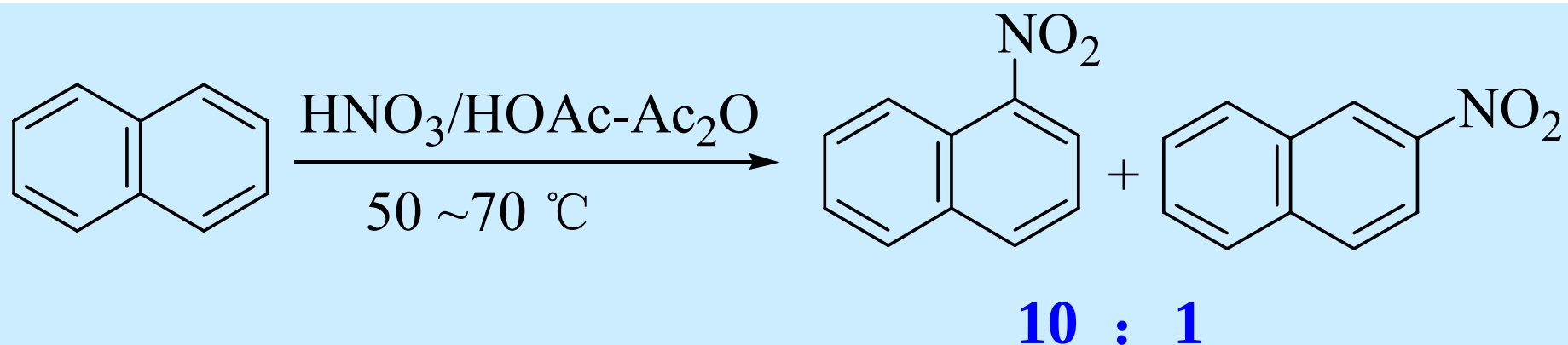
1. 萘 (Naphthalene)



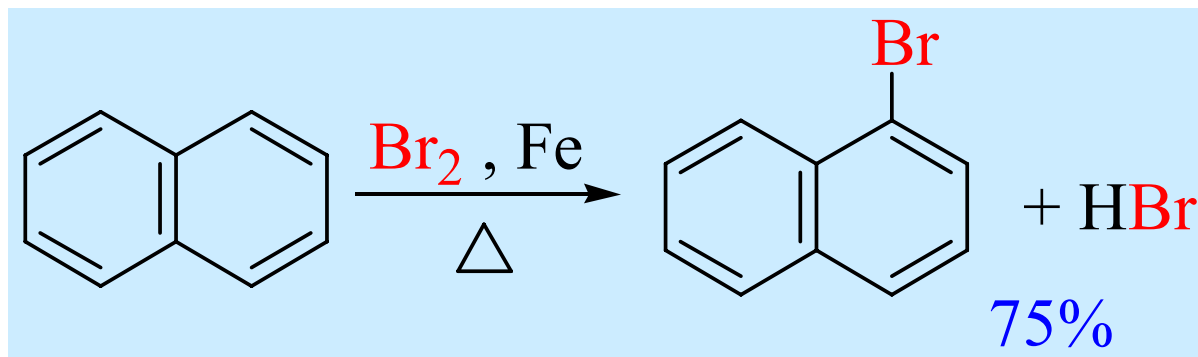
稳定化能 250 kJ/mol

2. 萘的反应—亲电取代反应

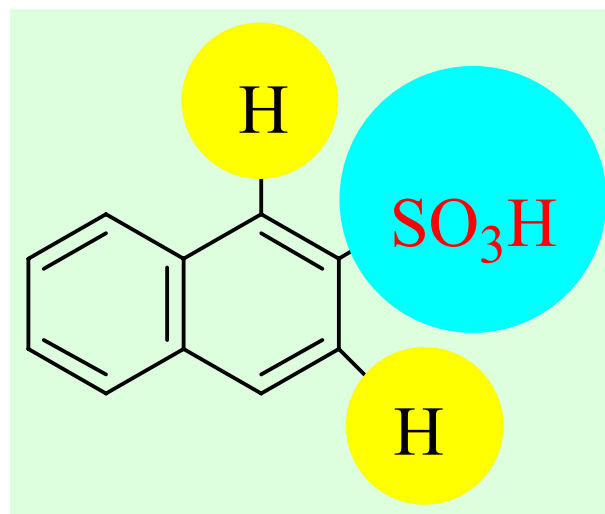
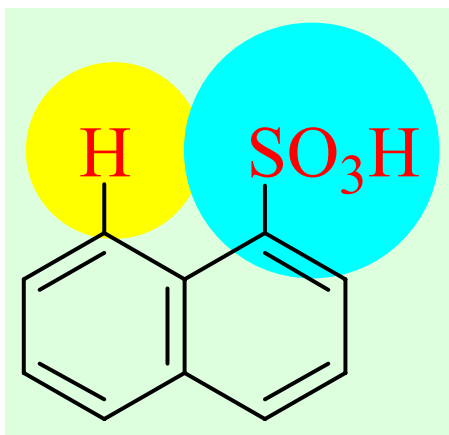
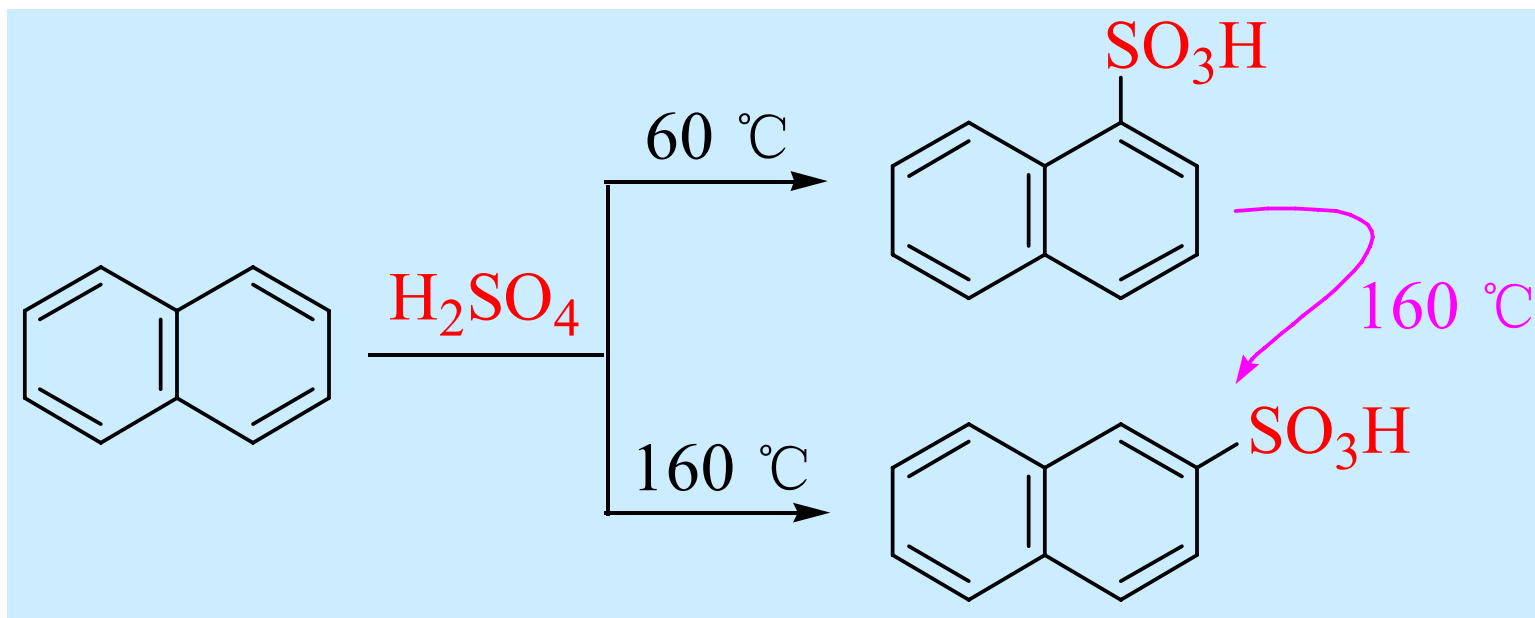
• 硝化



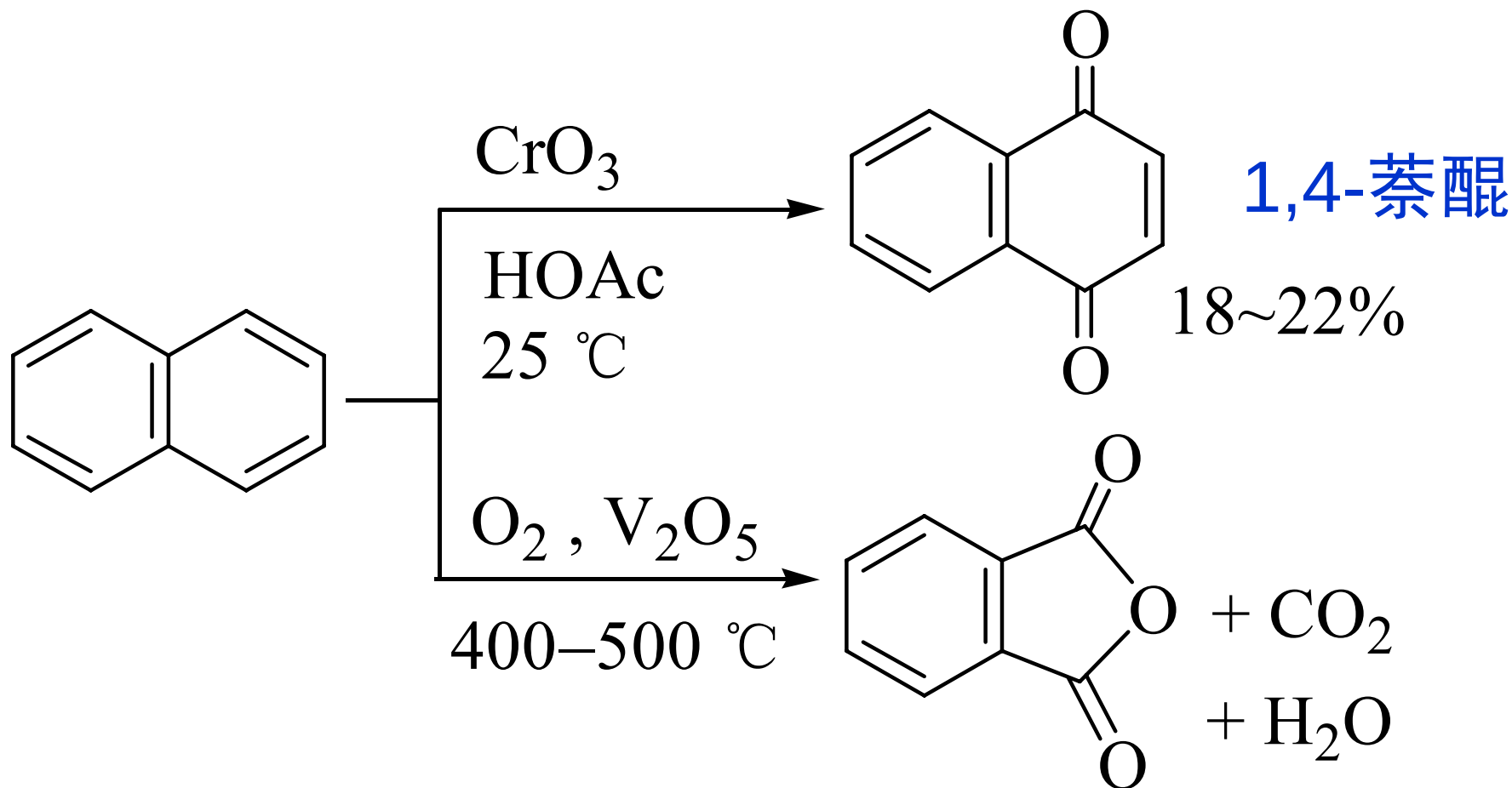
• 卤代



• 磺化

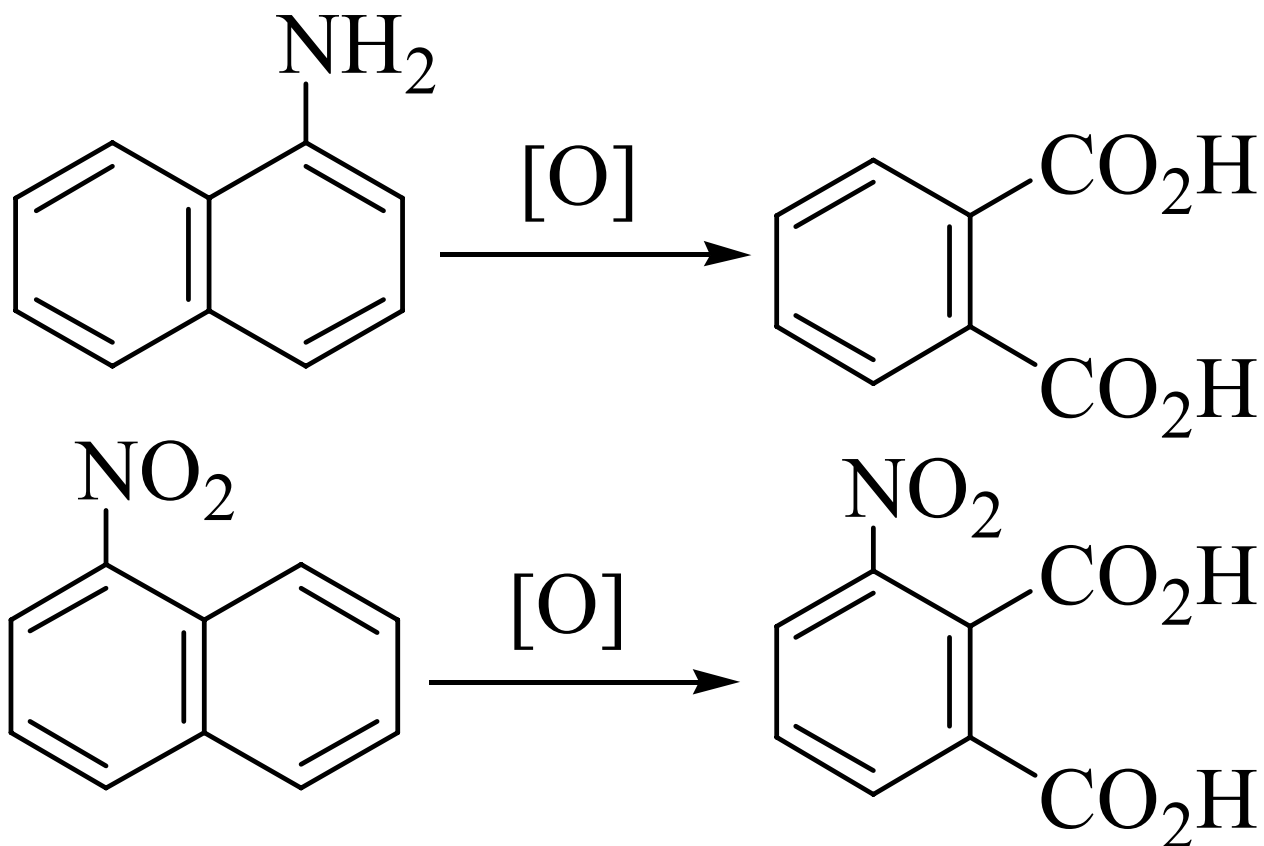


3. 萘的反应——氧化反应



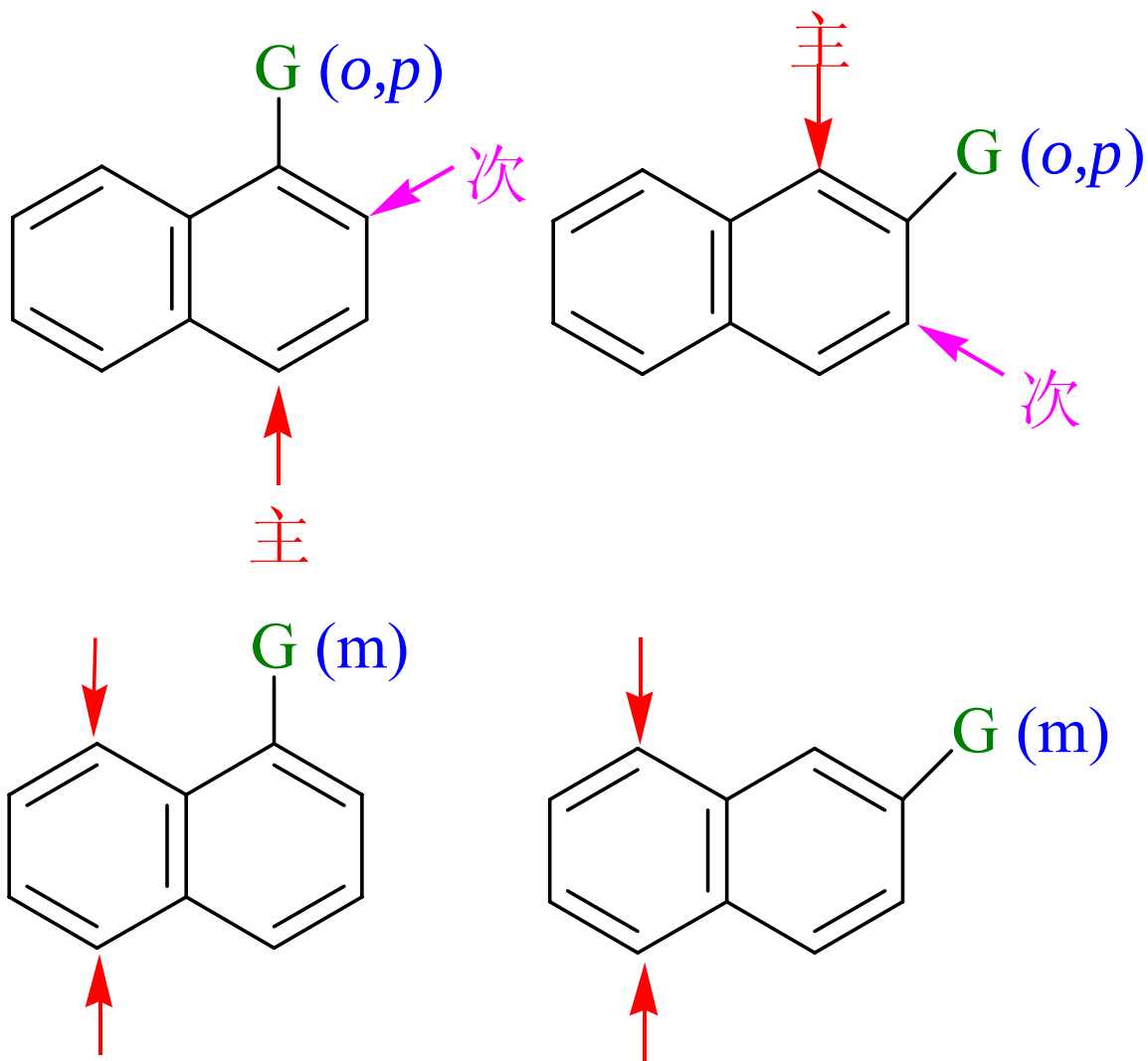
邻苯二甲酸酐（苯酐）

4. 取代萘的氧化反应



电子云密度较高的环被氧化

5. 取代萘亲电取代反应的定位



作业

P96: 6.3

P97: 6.4

P98: 6.5

P106: 6.9

P116: 2, 3, 4, 5, 7.